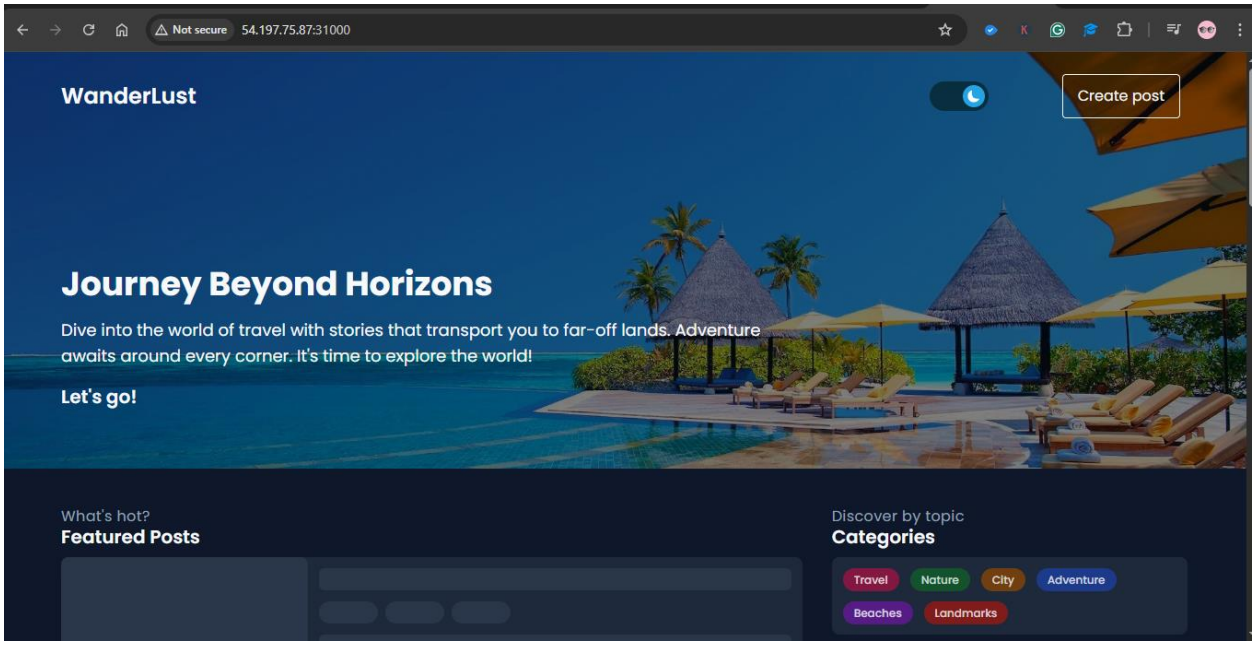
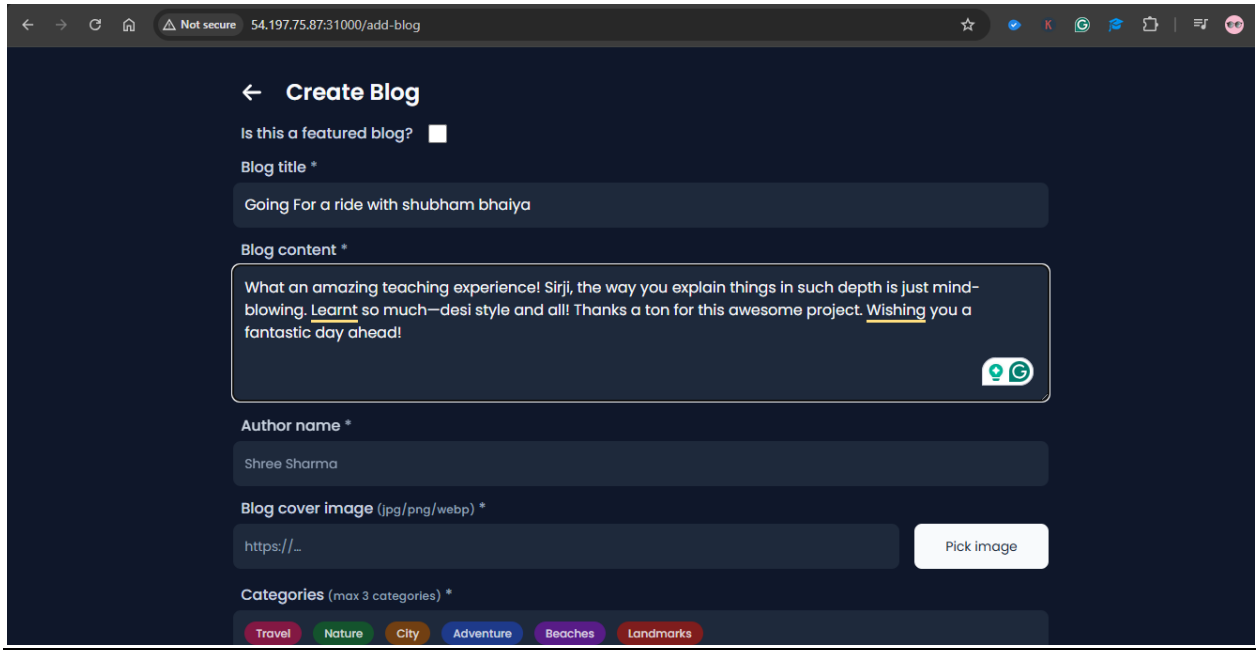
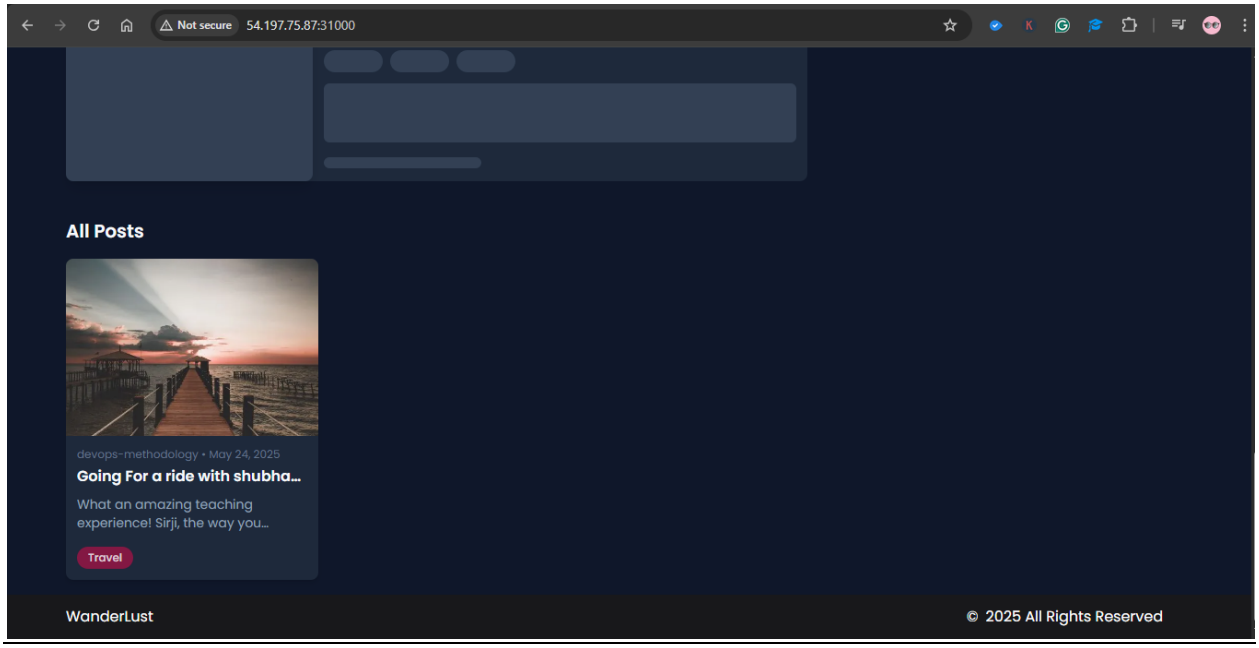
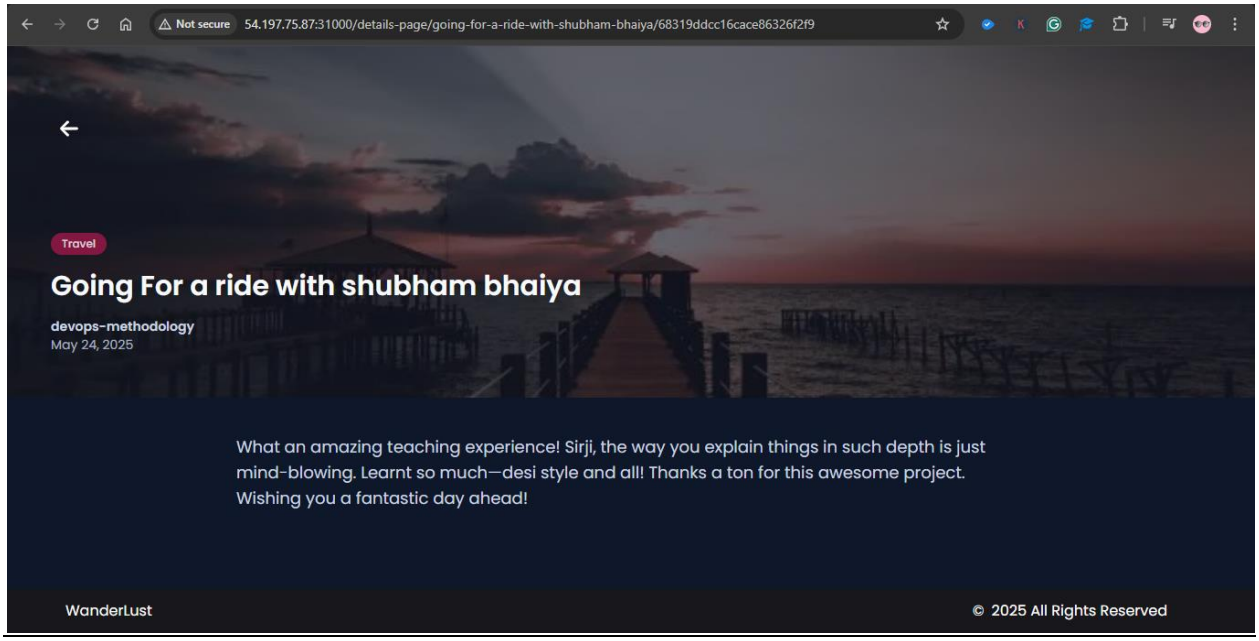
DEPLOYING A MERN STACK APPLICATION ON KUBERNETES USING KUBEADM | 3 TIER WEB APPLICATION





Devops-Methodology





Best wishes from Devops Methodology

Yay!! Backend of wanderlust app is now accessible

WHAT IS A MERN STACK ?

MERN uses four key technologies which used to build full stack web application.

M- mongo db – nosql database

E – express.js- backend web application framework

R-React.js- frontend java script library

N-node.js - javascript runtime (backend server)

This uses all technologies to build a web application from frontend to backend to database.

Architecture of mern stack app

React (Frontend)

|

Express.js + Node.js (Backend API)

|

MongoDB (Database)

What is 3 tier web application deployment?

Tier	Purpose	MERN Equivalent
Presentation	User interface (UI)	React.js
Application/Logic	Business logic, API handling	Express.js + Node.js
Data	Data storage and management	MongoDB

In this project we will do and travel blog application where we have to deploy the application but in this project we will deploy but with kubeadm(other options can be deployed like-minikube,eks,aks,k3s etc)

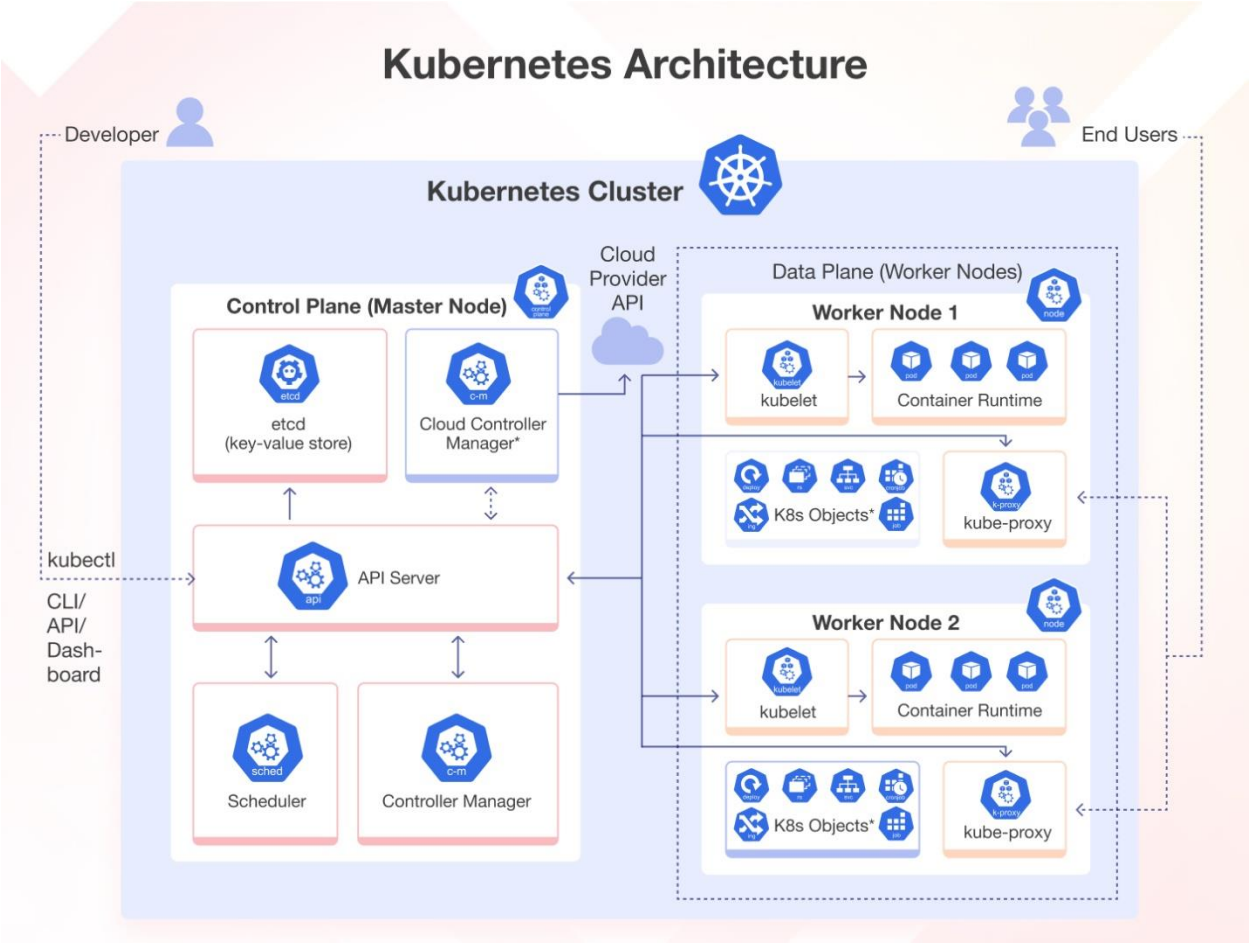
So to create a cluster how much minimum server we need..its about 2 server

1/master node(who will instruct to work)

2/worker node(who will do the real work accord to the instruction)

So we will launch 1/k8s-master

2/k8s-worker



Kubernetes architecture

In the master node/control plane we have got-

1/api server

2/controller manager

3/scheduler

4/etcd

Actually control plane is responsible for container orchestration and maintains the state of the cluster.

API SERVER:

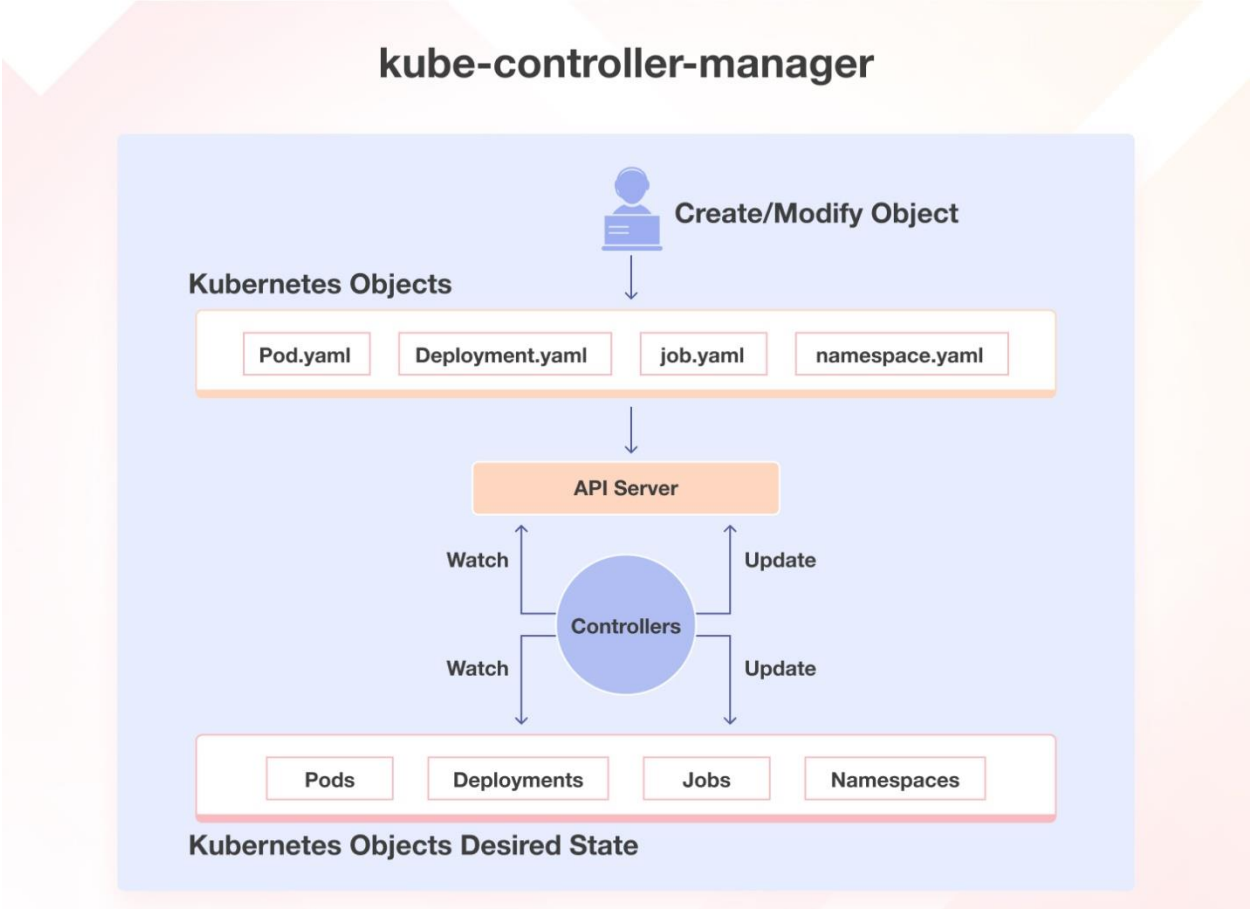
Here we as a devops engineer communicate with api server as cli command,api gateway or through kubernetes dashboard script.

Every thing inside the kubernetes or k8s is an api object.

Which we call through by a yaml script(yet another markup language)

Actually In short we can say api server is the entry point to the cluster.

Where as "**control manager"** maintains the state that desired state=actual state means how much pod we have defined it will manage the pod on the worker node.

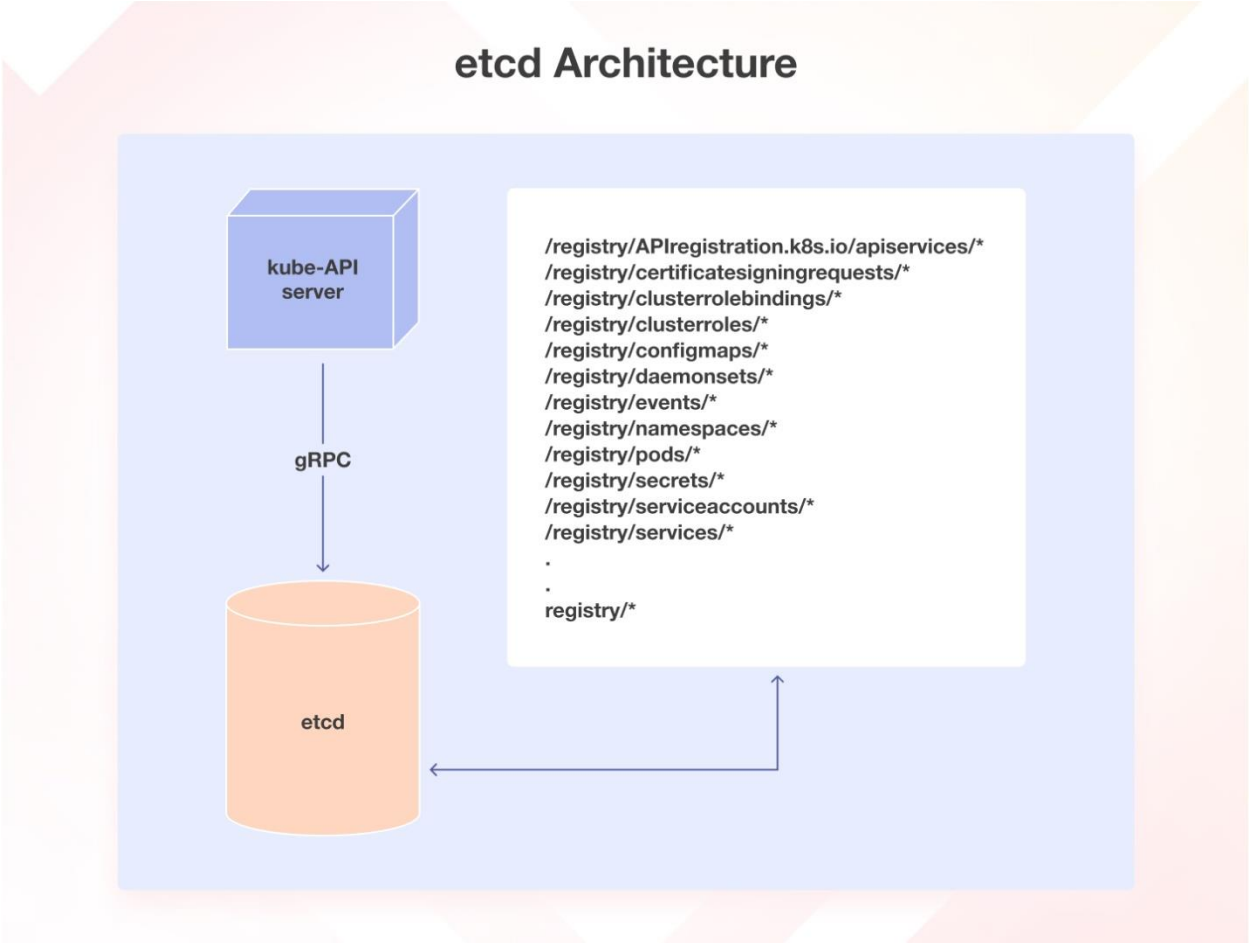


"Kube scheduler" –where as kube scheduler ensures the pod replacement(it does not replace the pod ,it only ensures it)…

User/devops engineer ----------Api--------control manager--------kube scheduler

How **kube-scheuler** know which pod to replace or which pod to create on which node(worker node)…

It gets the information from "**etcd "---**it stores the information in the key value pair accord to the pod requirements,cpu memory ,storage ,it schedules in the right pod.



It stores the data of only kubernetes cluster,such as information about pods,namespace, their state (etcd ------is only accessible to api server…)

api----- ------control manager---scheduler----etcd----worker node---

Worker Nodes

In this the real containerized application will work.

Mean there will 2 or more worker node contains pod inside the pod there will the container where our application will run…

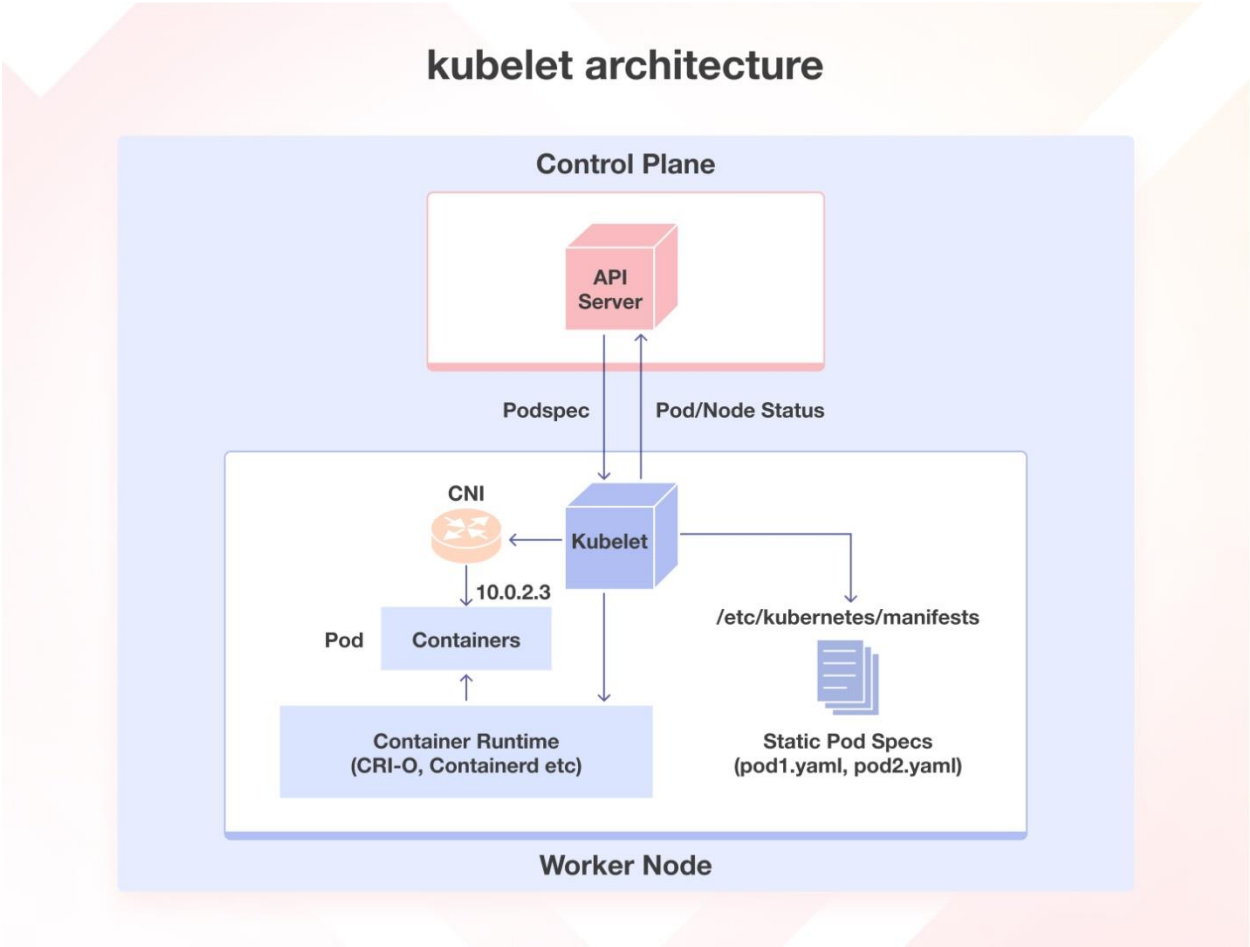
POD

It is the smallest unit of kubernetes,anything happen pod will be either terminated or destroyed,it is actually epiphermal in nature,it is abstraction over container,usually one application container inside the pod.

In the pod there will minimum one container if more than one container is there then it will be the side car container which will manage the applications by product/mainly it was there for application testing or other purpose may be.

KUBELET

In the worker node there will the api object that is **"kubelet"—**which is responsible for creation,deletion and modification of the container, accord to the requirement it also mount volumes by reading pod configuration which was called by api server in master node/control plane.

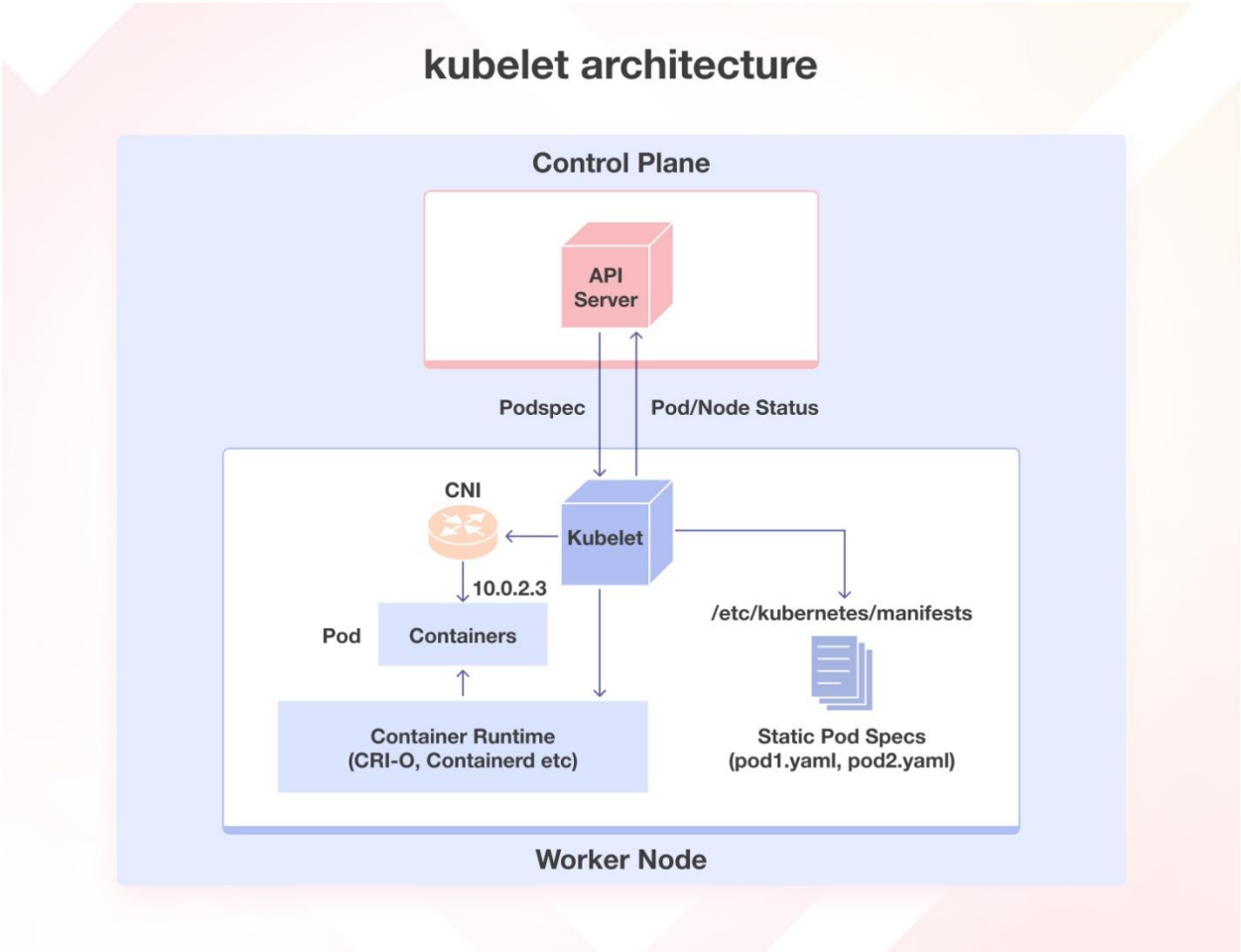


Actually the kubelet starts the api server,control manager and scheduler in the control plane..

Kubeproxy

Service objects expose to the ports for traffic and endpoint object contains pods IP address and ports,so the service objects works through this kubeproxy.

When you actually we use a service to expose pods,kubeproxy creates network rules for this service object to send the traffic to the pods,but it doesnot respond to http but runs on every node as a daemonset.



Actually it is responsible to create a master node and workernode to communicate with each other to form a robust system as a cluster.

Container Run Time(CNI):

It actually enables the total container lifecycle…like pulling images,allocation and isolation of the containers…

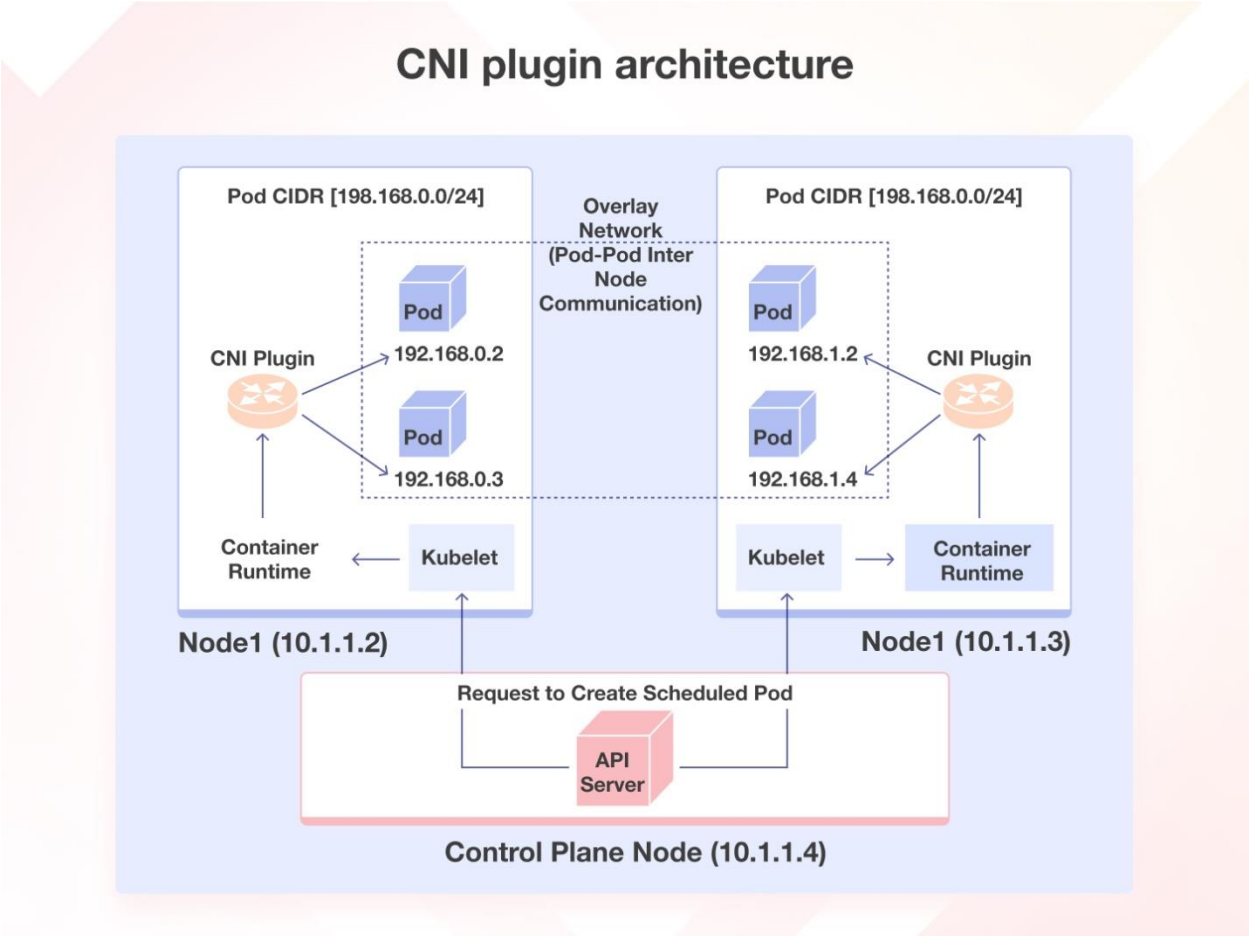
Kubelet-----CRT --(through CRI –container runtime interface)----which told the api for creating,deleting,starting and stopping the containers…by managing images and containers.

In this project in the master node we are installing CNI as calico for this purpose as a plugin.

```
# Network Plugin = calico
kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.26.0/manifests/calico.yaml

kubeadm token create --print-join-command
```

On Worker node

This is the calico plugin during installation of kubeadm which enables to create the control plane or master node.



PRACTICAL

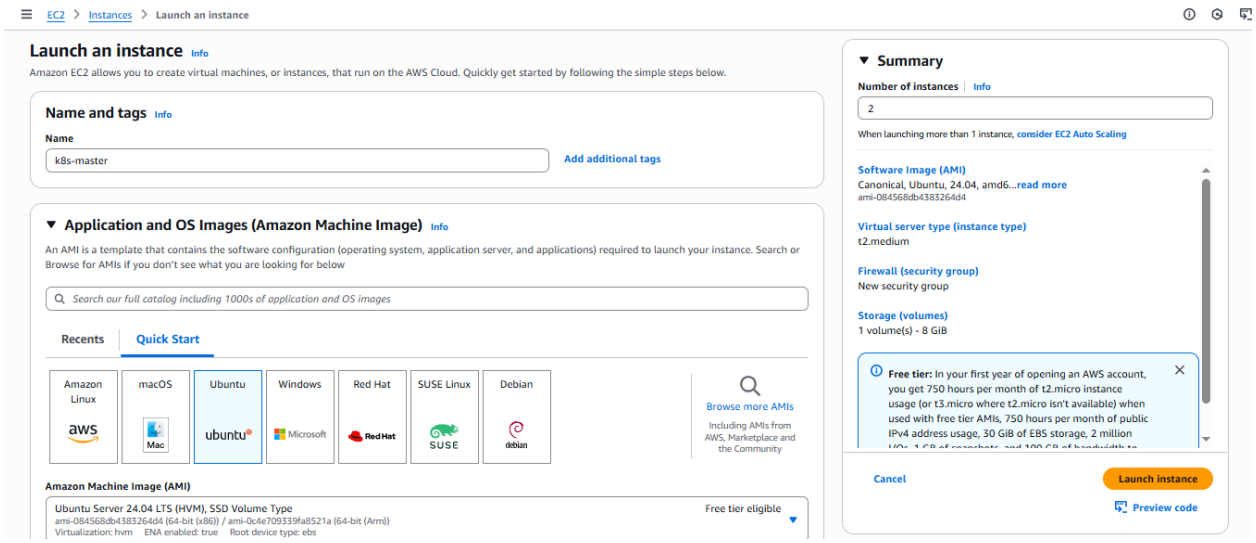
Here we will create two node 1.k8s-masternode

2.k8s workernode

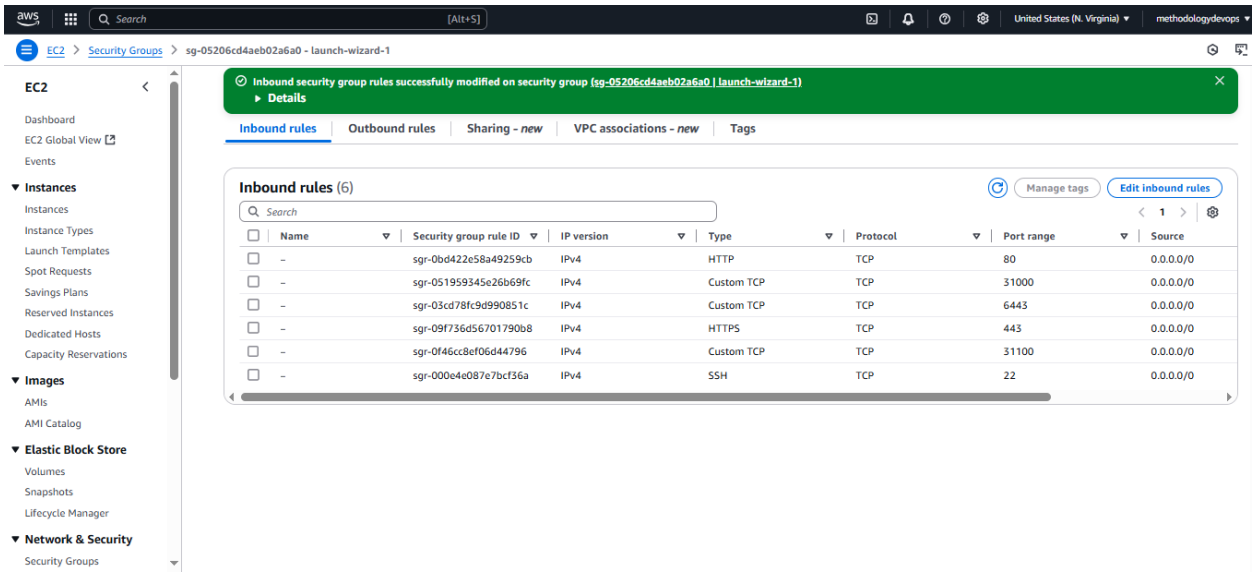
Ubuntu 24.04 lts

T2 medium

Storage 30gb



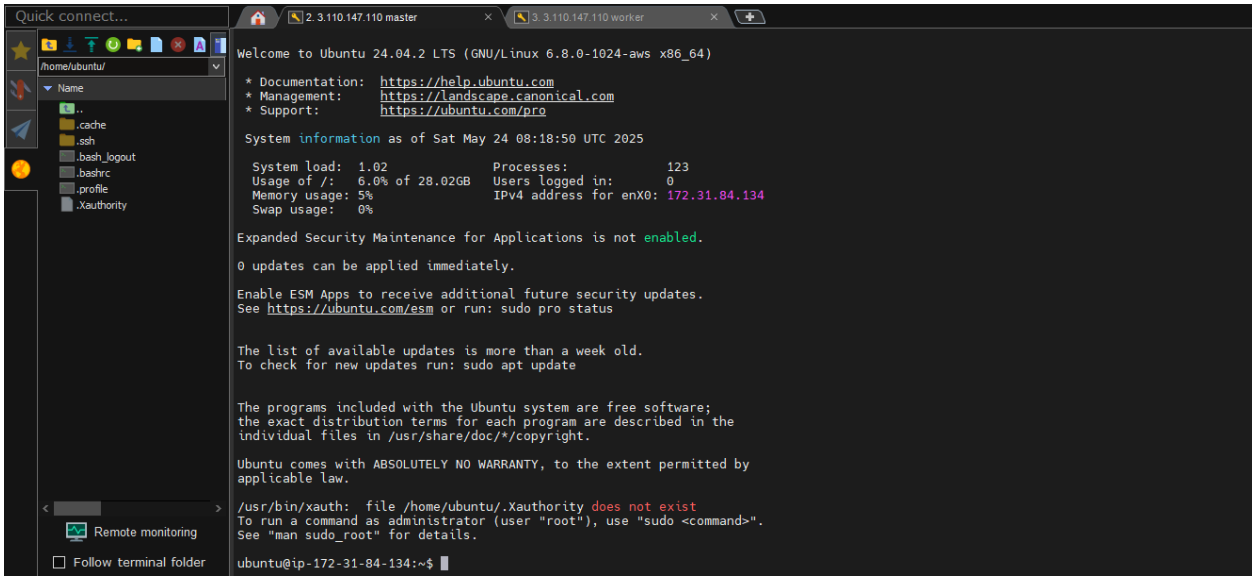
PORT TO BE OPENED



After that connect both with mobaxterm and in both nodes

> sudo su

>sudo apt update



Then add script1.sh in both nodes(which I below mentioned)

Script1.sh

```
#!/bin/bash

# disable swap

sudo swapoff -a


# Create the .conf file to load the modules at bootup

cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf

overlay

br_netfilter

EOF
```

```
sudo modprobe overlay

sudo modprobe br_netfilter


# sysctl params required by setup, params persist across reboots

cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf

net.bridge.bridge-nf-call-iptables  = 1

net.bridge.bridge-nf-call-ip6tables = 1

net.ipv4.ip_forward                 = 1

EOF


# Apply sysctl params without reboot

sudo sysctl --system


## Install CRIO Runtime

sudo apt-get update -y

sudo apt-get install -y software-properties-common curl apt-transport-https ca-certificates gpg


sudo curl -fsSL https://pkgs.k8s.io/addons:/cri-o:/prerelease:/main/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/cri-o-apt-keyring.gpg
```

```
echo "deb [signed-by=/etc/apt/keyrings/cri-o-apt-keyring.gpg] https://pkgs.k8s.io/addons:/cri-o:/prerelease:/main/deb/ /" | sudo tee /etc/apt/sources.list.d/cri-o.list


sudo apt-get update -y

sudo apt-get install -y cri-o


sudo systemctl daemon-reload

sudo systemctl enable crio --now

sudo systemctl start crio.service


echo "CRI runtime installed successfully"


# Add Kubernetes APT repository and install required packages

curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.29/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg

echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list


sudo apt-get update -y

sudo apt-get install -y kubelet="1.29.0-*" kubectl="1.29.0-*" kubeadm="1.29.0-*"

sudo apt-get update -y
```

sudo apt-get install -y jq

sudo systemctl enable --now kubelet

sudo systemctl start kubelet

---

then bash script1.sh in both nodes

what is in the script1.sh

- .

Then it installs ca-certificate and the gpg key for kubernetes and also install kubelet,kubectl and kubeadm

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# bash script1.sh
overlay
br_netfilter
net.bridge.bridge-nf-call-iptables  = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward                 = 1
* Applying /usr/lib/sysctl.d/10-apparmor.conf ...
* Applying /etc/sysctl.d/10-bufferbloat.conf ...
* Applying /etc/sysctl.d/10-console-messages.conf ...
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
* Applying /etc/sysctl.d/10-map-count.conf ...
* Applying /etc/sysctl.d/10-network-security.conf ...
* Applying /etc/sysctl.d/10-ptrace.conf ...
* Applying /etc/sysctl.d/10-zeropage.conf ...
* Applying /etc/sysctl.d/50-cloudimg-settings.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
* Applying /etc/sysctl.conf ...
kernel.apparmor_restrict_unprivileged_userns = 1
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
kernel.yama.ptrace_scope = 1
vm.mmap_min_addr = 65536
net.ipv4.neigh.default.gc_thresh2 = 15360
net.ipv4.neigh.default.gc_thresh3 = 16384
net.netfilter.nf_conntrack_max = 1048576
```

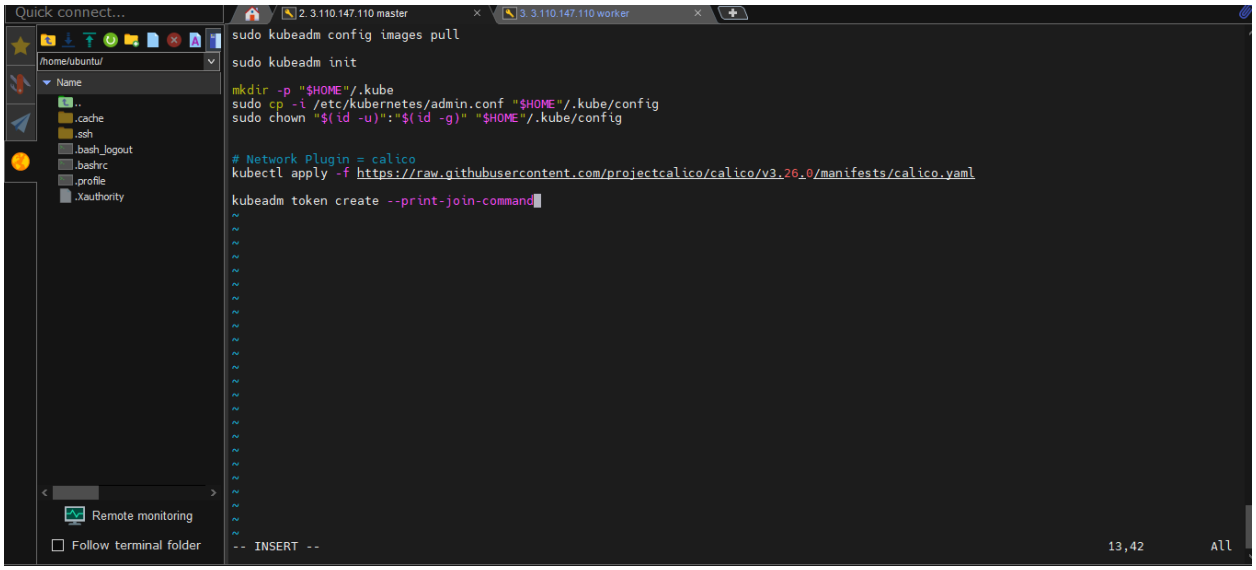
Then we have to copy and paste script2.sh on the master node

script2.h on master node

On Master node

- *Create a shell script 2.sh and paste the below code and run it*

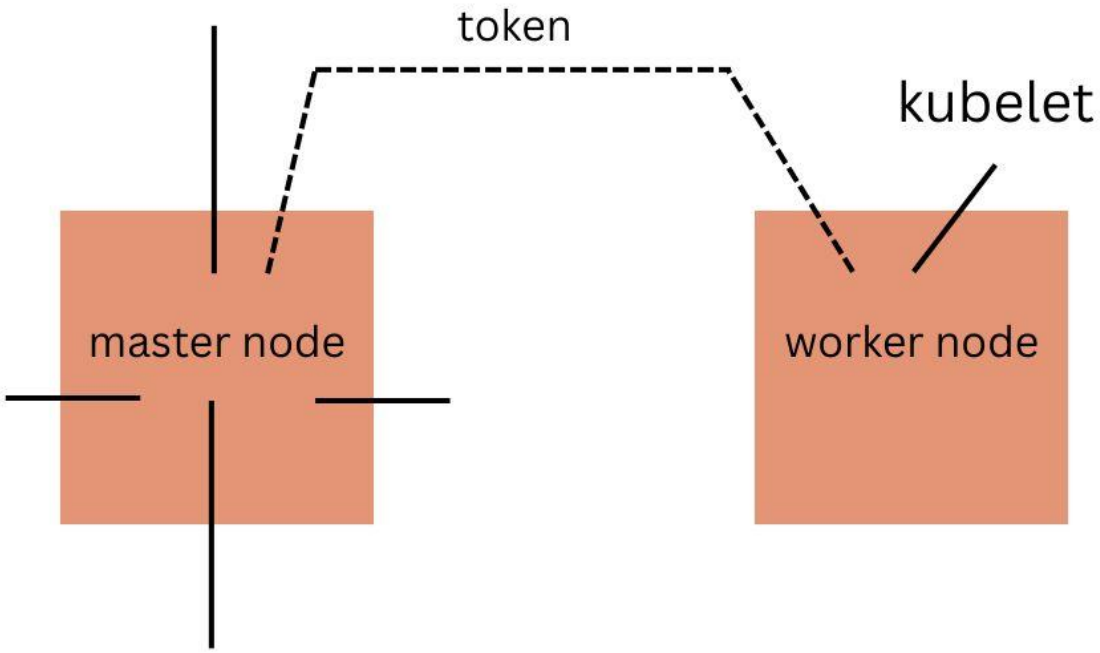
```
sudo kubeadm config images pull

sudo kubeadm init

mkdir -p "$HOME"/.kube
sudo cp -i /etc/kubernetes/admin.conf "$HOME"/.kube/config
sudo chown "$(id -u)":"$(id -g)" "$HOME"/.kube/config


# Network Plugin = calico
kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.26.0/manifests/calico.yaml

kubeadm token create --print-join-command
```



```
2. 3.110.147.110 master        3. 3.110.147.110 worker
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# vi script2.sh
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# bash script2.sh
I0524 08:30:49.242155    4185 version.go:256] remote version is much newer: v1.33.1; falling back to: stable-1.29
[config/images] Pulled registry.k8s.io/kube-apiserver:v1.29.15
```

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
Preparing to unpack .../7-kubeadm_1.29.0-1.1_amd64.deb ...
Unpacking kubeadm (1.29.0-1.1) ...
Setting up conntrack (1:1.4.8-1ubuntu1) ...
Setting up kubectl (1.29.0-1.1) ...
Setting up ebtables (2.0.11-6build1) ...
Setting up socat (1.8.0.0-4build3) ...
Setting up cri-tools (1.29.0-1.1) ...
Setting up kubernetes-cni (1.3.0-1.1) ...
Setting up kubelet (1.29.0-1.1) ...
Setting up kubeadm (1.29.0-1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/addons:/cri-o:/prerelease:/main/deb  InRelease
Hit:5 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.29/deb  InRelease
Hit:6 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3build1).
jq set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 74 not upgraded.
root@ip-172-31-93-124:/home/ubuntu# kubeadm join 172.31.84.134:6443 --token qpm7jy.uszipb0xwb5ef33w --discovery-token-ca-cert-hash sha256
:9588d27167b7d4082d95f50e7c8dacdb95867074d213b04c5a6611b880fe3b5f
```

After pasting that in last there will token generated in the master node which will connect master node to worker node



After that master node will connect with the worker node

You can check through kubectl get nodes in master node

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
I0524 08:34:12.171011    4303 kubeproxy.go:55] attempting to download the KubeProxyConfiguration from ConfigMap "kube-proxy"
I0524 08:34:12.174888    4303 kubelet.go:74] attempting to download the KubeletConfiguration from ConfigMap "kubelet-config"
I0524 08:34:12.178980    4303 initconfiguration.go:114] skip CRI socket detection, fill with the default CRI socket unix:///var/run/
inerd/containerd.sock
I0524 08:34:12.179162    4303 interface.go:432] Looking for default routes with IPv4 addresses
I0524 08:34:12.179185    4303 interface.go:437] Default route transits interface "enX0"
I0524 08:34:12.179287    4303 interface.go:209] Interface enX0 is up
I0524 08:34:12.179360    4303 interface.go:257] Interface "enX0" has 2 addresses :[172.31.93.124/20 fe80::10b0:6bff:fef6:b637/64].
I0524 08:34:12.179375    4303 interface.go:224] Checking addr  172.31.93.124/20.
I0524 08:34:12.179386    4303 interface.go:231] IP found 172.31.93.124
I0524 08:34:12.179404    4303 interface.go:263] Found valid IPv4 address 172.31.93.124 for interface "enX0".
I0524 08:34:12.179413    4303 interface.go:443] Found active IP 172.31.93.124
I0524 08:34:12.185440    4303 preflight.go:104] [preflight] Running configuration dependant checks
I0524 08:34:12.185468    4303 controlplaneprepare.go:225] [download-certs] Skipping certs download
I0524 08:34:12.185491    4303 kubelet.go:121] [kubelet-start] writing bootstrap kubelet config file at /etc/kubernetes/bootstrap-kub
conf
I0524 08:34:12.186254    4303 kubelet.go:136] [kubelet-start] writing CA certificate at /etc/kubernetes/pki/ca.crt
I0524 08:34:12.186941    4303 kubelet.go:157] [kubelet-start] Checking for an existing Node in the cluster with name "ip-172-31-93-1
nd status "Ready"
I0524 08:34:12.189417    4303 kubelet.go:172] [kubelet-start] Stopping the kubelet
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...
I0524 08:34:14.500459    4303 cert_rotation.go:137] Starting client certificate rotation controller
I0524 08:34:14.501074    4303 kubelet.go:220] [kubelet-start] preserving the crisocket information for the node
I0524 08:34:14.501099    4303 patchnode.go:31] [patchnode] Uploading the CRI Socket information "unix:///var/run/crio/crio.sock" to
ode API object "ip-172-31-93-124" as an annotation

This node has joined the cluster:
* Certificate signing request was sent to apiserver and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

root@ip-172-31-93-124:/home/ubuntu#
```

After that check the command as kubectl get nodes in master node

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get nodes
NAME               STATUS   ROLES           AGE     VERSION
ip-172-31-84-134   Ready    control-plane   3m19s   v1.29.0
ip-172-31-93-124   Ready    <none>          20s     v1.29.0
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

After that install docker in both nodes

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
Selecting previously unselected package dnsmasq-base.
Preparing to unpack .../5-dnsmasq-base_2.90-2build2_amd64.deb ...
Unpacking dnsmasq-base (2.90-2build2) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../6-docker.io_26.1.3-0ubuntu1~24.04.1_amd64.deb ...
Unpacking docker.io (26.1.3-0ubuntu1~24.04.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../7-ubuntu-fan_0.12.16_all.deb ...
Unpacking ubuntu-fan (0.12.16) ...
Setting up dnsmasq-base (2.90-2build2) ...
Setting up runc (1.1.12-0ubuntu3.1) ...
Setting up dns-root-data (2024071801~ubuntu0.24.04.1) ...
Setting up bridge-utils (1.7.1-1ubuntu2) ...
Setting up pigz (2.8-1) ...
Setting up containerd (1.7.24-0ubuntu1~24.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (26.1.3-0ubuntu1~24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `docker' (GID 113) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-93-124:/home/ubuntu# sudo chmod 777 /var/run/docker.sock
```

sudo apt install docker.io –y

for permission

sudo chmod 777 /var/run/docker.sock

after that go to master node

git clone –b devops https://github.com/devops-methodology/wanderlust.git

it will direct devops branch from the github

ls

cd backend/

ls –la

vim .env.docker

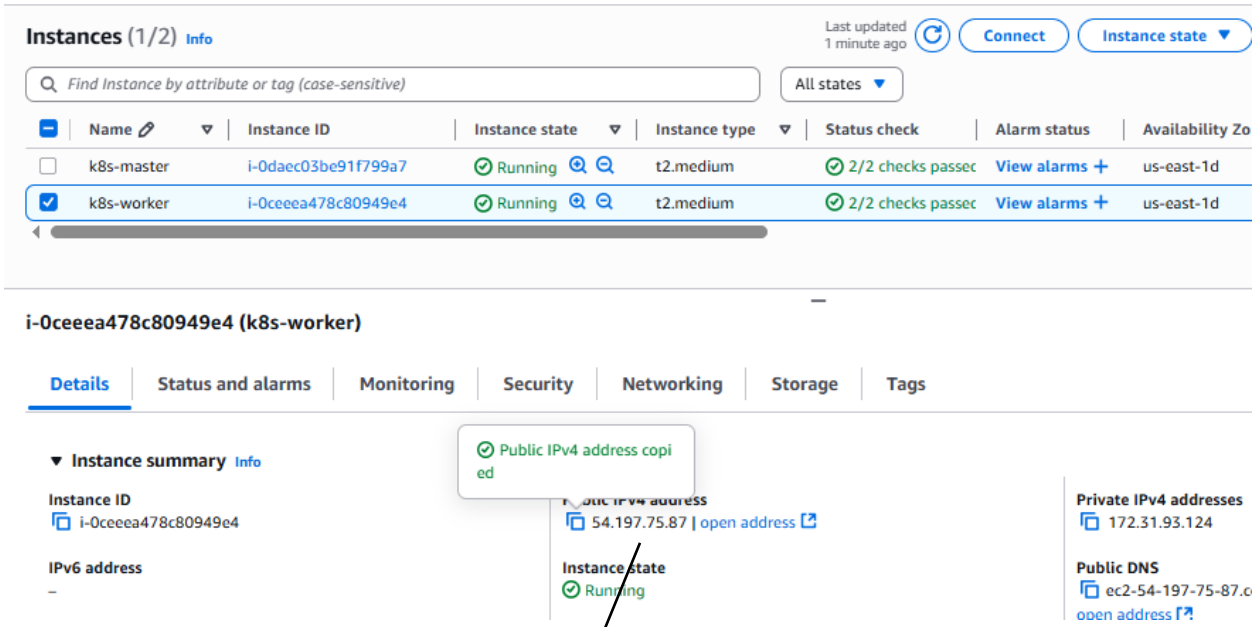
```
root@ip-172-31-84-134:/home/ubuntu/wanderlust# cd backend/
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# ls -la
total 420
drwxrwxr-x 12 ubuntu ubuntu   4096 May 24 08:22 .
drwxrwxr-x  9 ubuntu ubuntu   4096 May 24 08:22 ..
-rw-rw-r--  1 ubuntu ubuntu     38 May 24 08:22 .babelrc
-rw-rw-r--  1 ubuntu ubuntu    419 May 24 08:22 .env.docker
-rw-rw-r--  1 ubuntu ubuntu     72 May 24 08:22 .env.sample
-rw-rw-r--  1 ubuntu ubuntu     13 May 24 08:22 .eslintignore
-rw-rw-r--  1 ubuntu ubuntu    289 May 24 08:22 .eslintrc.json
-rw-rw-r--  1 ubuntu ubuntu     13 May 24 08:22 .prettierignore
-rw-rw-r--  1 ubuntu ubuntu    132 May 24 08:22 .prettierrc
-rw-rw-r--  1 ubuntu ubuntu    364 May 24 08:22 Dockerfile
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 api
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 config
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 controllers
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 data
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 models
-rw-rw-r--  1 ubuntu ubuntu 332414 May 24 08:22 package-lock.json
-rw-rw-r--  1 ubuntu ubuntu    956 May 24 08:22 package.json
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 public
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 routes
-rw-rw-r--  1 ubuntu ubuntu    914 May 24 08:22 server.js
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 services
drwxrwxr-x  5 ubuntu ubuntu   4096 May 24 08:22 tests
drwxrwxr-x  2 ubuntu ubuntu   4096 May 24 08:22 utils
-rw-rw-r--  1 ubuntu ubuntu    595 May 24 08:22 vercel.json
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# vim .env.docker
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

We have to give the k8s worker ip in the frontend url(because the frontend of the application will be work in workernode) then :wq

Devops-Methodology

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
MONGODB_URI="mongodb://mongo-service/wanderlust"
REDIS_URL="redis://redis-service:6379"
PORT=8080
FRONTEND_URL="http://13.52.243.88:5173"
ACCESS_COOKIE_MAXAGE=120000
ACCESS_TOKEN_EXPIRES_IN='120s'
REFRESH_COOKIE_MAXAGE=120000
REFRESH_TOKEN_EXPIRES_IN='120s'
JWT_SECRET=70dd8b38486eee723ce2505f6db06f1ee503fde5eb06fc04687191a0ed665f3f98776902d2c89f6b993b1c579a87fedaf584c693a106f7cbf16e8b4e67e9d6
df
NODE_ENV=Development
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
".env.docker" 10L, 419B                                                    1,1          All
```

Front end will work in workernode



Best wishes from Devops Methodology

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
MONGODB_URI="mongodb://mongo-service/wanderlust"
REDIS_URL="redis://redis-service:6379"
PORT=8080
FRONTEND_URL="http://54.197.75.87:5173"
ACCESS_COOKIE_MAXAGE=120000
ACCESS_TOKEN_EXPIRES_IN='120s'
REFRESH_COOKIE_MAXAGE=120000
REFRESH_TOKEN_EXPIRES_IN='120s'
JWT_SECRET=70dd8b38486eee723ce2505f6db06f1ee503fde5eb06fc04687191a0ed665f3f98776902d2c89f6b993b1c579a87fedaf584c693a106f7cbf16e8b4e67e9d6
df
NODE_ENV=Development
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                    4,35          All
```

After that we have to create a namespace named as wanderlust

Why we are creating namespace

Because in a cluster a lot of resources will create we will confuse whom to export whom to work with so we can't organize the things,so that's why we are going to create folder in cluster so that the resources with in the group will not overlap with each other and will be in isolation process for which we can easily work accord to the services needed(it will group the resources)

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# kubectl create ns wanderlust
namespace/wanderlust created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

After that check

Kubectl create ns wanderlust

Kubectl get ns

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# kubectl get ns
NAME              STATUS   AGE
default           Active   19m
kube-node-lease   Active   19m
kube-public       Active   19m
kube-system       Active   19m
wanderlust        Active   74s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

We will see there is a default name space

We have to write another command

kubectl config set-context --current --namespace wanderlust

why we are doing this as we don't want to write repeatedly –n namespace during any kubectl get nodes,pods or logs if not in every command we have to write –n namespace which is not a good practice.

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# kubectl config set-context --current --namespace wanderlust
Context "kubernetes-admin@kubernetes" modified.
```

Then write the command as

kubectl get pods -n kube-system -o wide | grep -i core

why we are write this command we can check that the dns is working on the master node only

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# kubectl get pods -n kube-system -o wide | grep -i core
coredns-76f75df574-8rw25   1/1     Running   0          22m   192.168.135.67   ip-172-31-84-134   <none>           <none>
coredns-76f75df574-gk55s   1/1     Running   0          22m   192.168.135.66   ip-172-31-84-134   <none>           <none>
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

The core dns is in master node so we have edit in core dns because and replace

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
  creationTimestamp: "2025-05-24T08:31:17Z"
  generation: 1
  labels:
    k8s-app: kube-dns
  name: coredns
  namespace: kube-system
  resourceVersion: "606"
  uid: dddf2f11-a86e-4b63-a211-77ce5018242e
spec:
  progressDeadlineSeconds: 600
  replicas: 4
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      k8s-app: kube-dns
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 1
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        k8s-app: kube-dns
    spec:
      affinity:
        podAntiAffinity:
-- INSERT --                                                              20,14         Top
```

Replace the replica as 4

So that dns will resolve in worker node also

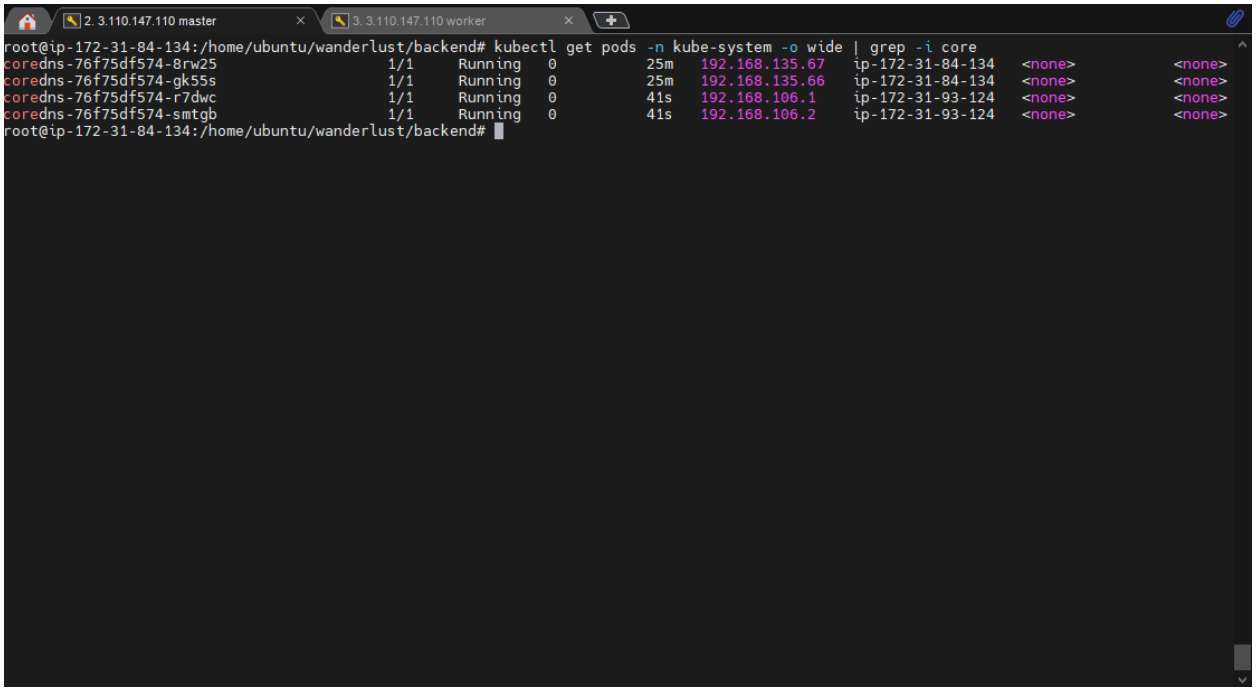
```
2. 3.110.147.110 master        3. 3.110.147.110 worker
      schedulerName: default-scheduler
      securityContext: {}
      serviceAccount: coredns
      serviceAccountName: coredns
      terminationGracePeriodSeconds: 30
      tolerations:
      - key: CriticalAddonsOnly
        operator: Exists
      - effect: NoSchedule
        key: node-role.kubernetes.io/control-plane
      volumes:
      - configMap:
          defaultMode: 420
          items:
          - key: Corefile
            path: Corefile
          name: coredns
        name: config-volume
status:
  availableReplicas: 2
  conditions:
  - lastTransitionTime: "2025-05-24T08:31:49Z"
    lastUpdateTime: "2025-05-24T08:31:49Z"
    message: Deployment has minimum availability.
    reason: MinimumReplicasAvailable
    status: "True"
    type: Available
  - lastTransitionTime: "2025-05-24T08:31:22Z"
    lastUpdateTime: "2025-05-24T08:31:52Z"
    message: ReplicaSet "coredns-76f75df574" has successfully progressed.
    reason: NewReplicaSetAvailable
    status: "True"
    type: Progressing
  observedGeneration: 1
  readyReplicas: 2
  replicas: 2
  updatedReplicas: 2
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

You can check now the dns is pointing to both worker node and master node

DNS translates human-friendly domain names (like `google.com`) into IP addresses (like `142.250.195.78`) so that browsers and apps can load websites and services.

Instead of remembering numbers, we use names — and DNS handles the behind-the-scenes conversion.

So mongodb –service can know about master node and worker node as mongodb-service is exposed to the application port.



```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# ls
Dockerfile  config       data    package-lock.json  public  server.js  tests  vercel.json
api         controllers  models  package.json       routes  services   utils
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# cat Dockerfile
# Stage 1
FROM node:21 AS backend-builder

# setup the working dir
WORKDIR /app

# code
COPY . .

# packages install
RUN npm i

# tests
RUN npm run test

# Stage 2
FROM node:21-slim

# setup the working dir
WORKDIR /app

# copy the above stage as compressed
COPY --from=backend-builder /app .

COPY .env.docker .env

# Port
EXPOSE 8080

# App
CMD ["npm", "start"]
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

Then go to the backend vi .env.dock

There also change the ip and copy of the workernode ip and paste in that(it's the environment variables we are changing because accord to that system will work)

```
2. 3.110.147.110 master
VITE_API_PATH="http://54.197.75.87:31100"
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

Then we have to build the image and push it to docker hub and then there k8s will pull the image and after the deployment it will create the container that is application and we can see our wanderlust website.

docker build -t premd9/frontend-wanderlust:latest .

docker build –t premd9/backend-wanderlust:latest .

Devops-Methodology

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# ls
Dockerfile  config       data    package-lock.json  public  server.js  tests  vercel.json
api         controllers  models  package.json       routes  services   utils
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# cd ..
root@ip-172-31-84-134:/home/ubuntu/wanderlust# cd frontend/
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# l
Dockerfile  components.json  jest.config.ts     package.json       public/  tailwind.config.js  tsconfig.node.json  verce
README.md   index.html       package-lock.json  postcss.config.js  src/     tsconfig.json       tsconfig.prod.json  vite
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# ls
Dockerfile  components.json  jest.config.ts     package.json       public  tailwind.config.js  tsconfig.node.json  vercel
README.md   index.html       package-lock.json  postcss.config.js  src     tsconfig.json       tsconfig.prod.json  vite.c
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# vi .env.docker
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# docker build -t premd91/frontend-wanderlust:latest.
```

```
README.md   index.html       package-lock.json  postcss.config.js  src     tsconfig.json       tsconfig.prod.json  vite.confi
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# vi .env.docker
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend# docker build -t premd91/frontend-wanderlust:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon    918kB
Step 1/11 : FROM node:21 AS frontend-builder
21: Pulling from library/node
c6cf28de8a06: Extracting [==================================================>]  48.23MB/49.58MB
891494355808: Download complete
6582c62583ef: Download complete
bf2c3e352f3d: Download complete
2b542796ccab: Download complete
4684276225df: Download complete
08750d53428e: Download complete
5d3106ce01c3: Download complete
```

Best wishes from Devops Methodology

Devops-Methodology

```
 ---> Removed intermediate container 0fc757d9d997
 ---> 59445f6cf769
Step 5/11 : COPY . .
 ---> 4597e5157560
Step 6/11 : FROM node:21-slim
21-slim: Pulling from library/node
09f376ebb190: Pull complete
44f5fe9df22b: Pull complete
9f1c458e7f2d: Pull complete
2e82ac84e151: Pull complete
30e51537f486: Pull complete
Digest: sha256:dfc05dee209a1d7adf2ef189bd97396daad4e97c6eaa85778d6f75205ba1b0fb
Status: Downloaded newer image for node:21-slim
 ---> 539323daf922
Step 7/11 : WORKDIR /app
 ---> Running in 6ff758730b18
 ---> Removed intermediate container 6ff758730b18
 ---> eb9e34eb125f
Step 8/11 : COPY --from=frontend-builder /app .
 ---> 91fa4bae73b4
Step 9/11 : COPY .env.docker .env.local
 ---> d2f1e62edb6b
Step 10/11 : EXPOSE 5173
 ---> Running in b02dfe80b49d
 ---> Removed intermediate container b02dfe80b49d
 ---> 9b239b04f0e5
Step 11/11 : CMD ["npm", "run", "dev", "--", "--host"]
 ---> Running in b6306ee3e530
 ---> Removed intermediate container b6306ee3e530
 ---> 5819db2d698c
Successfully built 5819db2d698c
Successfully tagged premd91/frontend-wanderlust:latest
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend#
```

```
REPOSITORY                  TAG       IMAGE ID       CREATED              SIZE
premd91/frontend-wanderlust latest    5819db2d698c   53 seconds ago       491MB
<none>                      <none>    4597e5157560   About a minute ago   1.48GB
node                        21-slim   539323daf922   12 months ago        203MB
node                        21        5c8697e82213   12 months ago        1.1GB
root@ip-172-31-84-134:/home/ubuntu/wanderlust/frontend#
```

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# ls
Dockerfile  config       data    package-lock.json  public  server.js  tests  vercel.json
api         controllers  models  package.json       routes  services   utils
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker build -t premd91/backend-wanderlust:latest .
```

Best wishes from Devops Methodology

```
2. 3.110.147.110 master        3. 3.110.147.110 worker
 app                    |   93.33 |      100 |       50 |   93.33 |
  server.js             |   93.33 |      100 |       50 |   93.33 | 30
 app/config             |   78.57 |      100 |    66.66 |   78.57 |
  db.js                 |      70 |      100 |    66.66 |      70 | 7-8,14
  utils.js              |     100 |      100 |      100 |     100 |
 app/controllers        |   38.09 |    34.54 |    47.05 |   38.09 |
  auth-controller.js    |    8.59 |        0 |        0 |    8.59 | 13-43,52-94,104-113,118-172,181-232,242-251,256-312,321-370,379-385
  posts-controller.js   |     100 |       95 |      100 |     100 | 16
 app/models             |     100 |      100 |      100 |     100 |
  post.js               |     100 |      100 |      100 |     100 |
  user.js               |     100 |      100 |      100 |     100 |
 app/routes             |     100 |      100 |      100 |     100 |
  auth.js               |     100 |      100 |      100 |     100 |
  posts.js              |     100 |      100 |      100 |     100 |
 app/services           |    62.5 |       50 |      100 |    62.5 |
  redis.js              |    62.5 |       50 |      100 |    62.5 | 9-13,18
 app/tests/utils        |     100 |      100 |      100 |     100 |
  helper-objects.js     |     100 |      100 |      100 |     100 |
 app/utils              |    62.5 |       40 |      100 |   57.14 |
  cache-posts.js        |   43.75 |     37.5 |      100 |   30.76 | 12-17,23-24,30-31
  constants.js          |     100 |      100 |      100 |     100 |
  cookie_options.js     |     100 |      100 |      100 |     100 |
  middleware.js         |   66.66 |       50 |      100 |    62.5 | 8-9,13
----------------------|---------|----------|---------|---------|-------------------
Test Suites: 1 failed, 1 passed, 2 total
Tests:       9 failed, 28 passed, 37 total
Snapshots:   0 total
Time:        50.202 s
Ran all test suites.
 ---> Removed intermediate container 5344a2c968ac
 ---> 81d9e7215679
Step 6/11 : FROM node:21-slim
 ---> 539323daf922
Step 7/11 : WORKDIR /app
 ---> Using cache
 ---> eb9e34eb125f
Step 8/11 : COPY --from=backend-builder /app .
```

```
 ---> Removed intermediate container 5344a2c968ac
 ---> 81d9e7215679
Step 6/11 : FROM node:21-slim
 ---> 539323daf922
Step 7/11 : WORKDIR /app
 ---> Using cache
 ---> eb9e34eb125f
Step 8/11 : COPY --from=backend-builder /app .
 ---> 59bed0f41c5c
Step 9/11 : COPY .env.docker .env
 ---> 53391c1bad2b
Step 10/11 : EXPOSE 8080
 ---> Running in 2f058f0b2824
 ---> Removed intermediate container 2f058f0b2824
 ---> dd30027a6832
Step 11/11 : CMD ["npm", "start"]
 ---> Running in 80acfc9c944e
 ---> Removed intermediate container 80acfc9c944e
 ---> b8e8255d71b8
Successfully built b8e8255d71b8
Successfully tagged premd91/backend-wanderlust:latest
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

You can check by the command

docker images

```
Successfully built b8e8255d71b8
Successfully tagged premd91/backend-wanderlust:latest
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker images
REPOSITORY                    TAG       IMAGE ID       CREATED          SIZE
premd91/backend-wanderlust    latest    b8e8255d71b8   30 seconds ago   273MB
<none>                        <none>    81d9e7215679   43 seconds ago   1.2GB
premd91/frontend-wanderlust   latest    5819db2d698c   5 minutes ago    491MB
<none>                        <none>    4597e5157560   5 minutes ago    1.48GB
node                          21-slim   539323daf922   12 months ago    203MB
node                          21        5c8697e82213   12 months ago    1.1GB
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

Then we have to push

Before that create “pat” which is the authorization token and will be generated by going to dockerhug>account>security>generate token

After that docker push

```
2. 3.110.147.110 master    3. 3.110.147.110 worker
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker login -u premd91
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

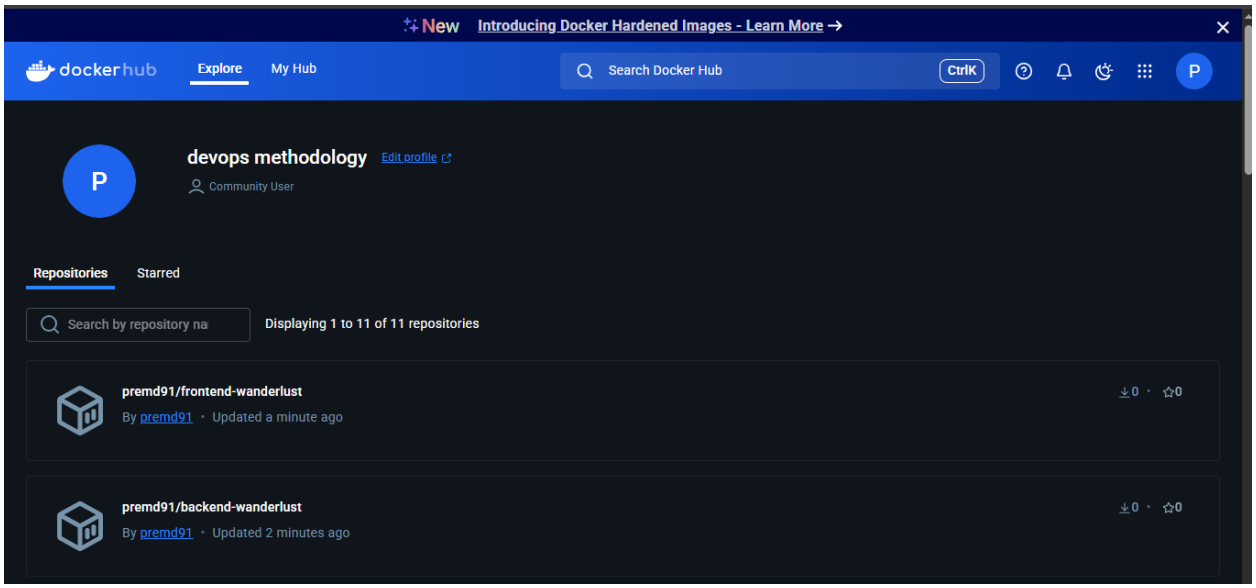
Then docker push backend and frontend

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker push premd91/backend-wanderlust:latest
The push refers to repository [docker.io/premd91/backend-wanderlust]
35a04a960f80: Pushed
c1400be8bdf5: Pushing [==================================>              ]  51.16MB/70.53MB
f2a8087c0e2f: Pushed
bbef00473d3a: Mounted from library/node
9bf32ad45c66: Mounted from library/node
3281047762f0: Mounted from library/node
6f611743c2e1: Mounted from library/node
5d4427064ecc: Mounted from library/node
```

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker push premd91/frontend-wanderlust:latest
The push refers to repository [docker.io/premd91/frontend-wanderlust]
d5ded2d4aba9: Pushed

f2a8087c0e2f: Mounted from premd91/backend-wanderlust
bbef00473d3a: Mounted from premd91/backend-wanderlust
9bf32ad45c66: Mounted from premd91/backend-wanderlust
3281047762f0: Mounted from premd91/backend-wanderlust
6f611743c2e1: Mounted from premd91/backend-wanderlust
5d4427064ecc: Mounted from premd91/backend-wanderlust
```

```
The push refers to repository [docker.io/premd91/backend-wanderlust]
85a04a960f80: Pushed
c1400be8bdf5: Pushed
f2a8087c0e2f: Pushed
bbef00473d3a: Mounted from library/node
9bf32ad45c66: Mounted from library/node
3281047762f0: Mounted from library/node
6f611743c2e1: Mounted from library/node
5d4427064ecc: Mounted from library/node
latest: digest: sha256:a7628e36a01a09998bd276b6a27c7df7567536dff304ec275cde13d34b4b0065 size: 1992
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend# docker push premd91/frontend-wanderlust:latest
The push refers to repository [docker.io/premd91/frontend-wanderlust]
d5ded2d4aba9: Pushed
12063f67bbbe: Pushed
f2a8087c0e2f: Mounted from premd91/backend-wanderlust
bbef00473d3a: Mounted from premd91/backend-wanderlust
9bf32ad45c66: Mounted from premd91/backend-wanderlust
3281047762f0: Mounted from premd91/backend-wanderlust
6f611743c2e1: Mounted from premd91/backend-wanderlust
5d4427064ecc: Mounted from premd91/backend-wanderlust
latest: digest: sha256:587999cf39470dd97738ac1340c481cb206a17e8ac4d6366f87764e539a0990b size: 1992
root@ip-172-31-84-134:/home/ubuntu/wanderlust/backend#
```

You can check in the docker hub both for backend and frontend



Next we have to deploy it

cd ../kubernetes

kubectl apply -f persistentVolume.yaml(because we have to create volume for the deployment)

then storage we want we have to claim it

so we will create persistent volume claim

kubectl apply -f persistentVolumeClaim.yaml

kubectl apply -f mongodb.yaml

kubectl apply -f redis.yaml

Before that in backend.yaml and frontend.yaml

We have to change the docker image which we have pushed to dockerhub

kubectl apply -f backend.yaml

kubectl apply -f frontend.yaml

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# ls
README.md  backend.yaml   kubeadm.md    persistentVolume.yaml       redis.yaml  script2.sh
assets     frontend.yaml  mongodb.yaml  persistentVolumeClaim.yaml  script1.sh
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f persistentVolume.yaml
persistentvolume/mongo-pv created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f persistentVolumeClaim.yaml
persistentvolumeclaim/mongo-pvc created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f mongodb.yaml
deployment.apps/mongo-deployment created
service/mongo-service created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

Then we can check by

Kubectl get pods

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mongo-deployment-9d8749988-nghgn    1/1     Running   0          24s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get pods -n wanderlust
NAME                                READY   STATUS    RESTARTS   AGE
mongo-deployment-9d8749988-nghgn    1/1     Running   0          34s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# ls
README.md  backend.yaml   kubeadm.md    persistentVolume.yaml       redis.yaml  script2.sh
assets     frontend.yaml  mongodb.yaml  persistentVolumeClaim.yaml  script1.sh
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f persistentVolume.yaml
persistentvolume/mongo-pv created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f persistentVolumeClaim.yaml
persistentvolumeclaim/mongo-pvc created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f mongodb.yaml
deployment.apps/mongo-deployment created
service/mongo-service created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mongo-deployment-9d8749988-nghgn    1/1     Running   0          24s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get pods -n wanderlust
NAME                                READY   STATUS    RESTARTS   AGE
mongo-deployment-9d8749988-nghgn    1/1     Running   0          34s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f redis.yaml
deployment.apps/redis-deployment created
service/redis-service created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mongo-deployment-9d8749988-nghgn    1/1     Running   0          2m7s
redis-deployment-9fbf97b49-xxmjk    1/1     Running   0          5s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

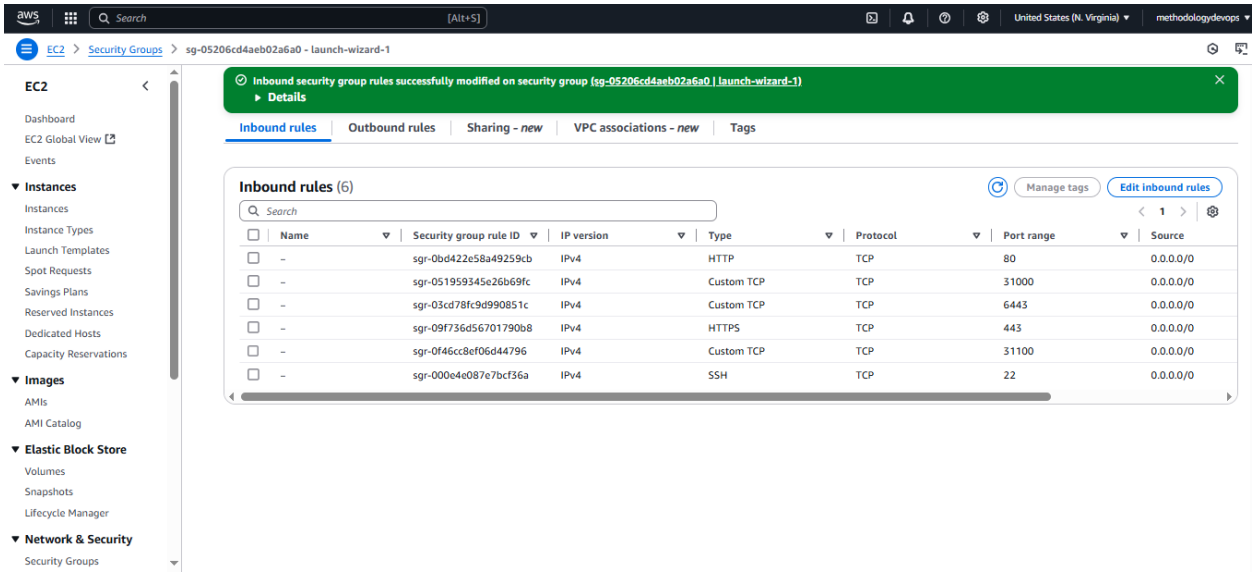
2. 3.110.147.110 master | 3. 3.110.147.110 worker

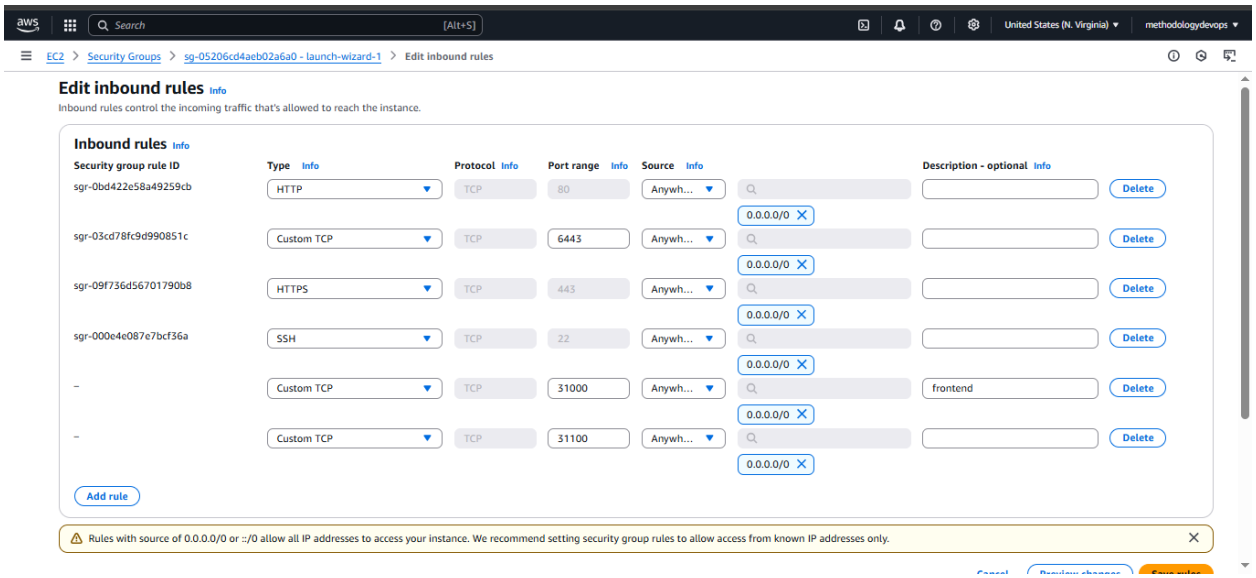
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deployment
  namespace: wanderlust
  labels:
    app: frontend
spec:
  replicas: 1
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      name: frontend
      labels:
        app: frontend
    spec:
      containers:
        - name: frontend
          image: premd91/wanderlust-frontend-beta:latest
          ports:
            - containerPort: 5173
---
apiVersion: v1
kind: Service
metadata:
  name: frontend-service
  namespace: wanderlust
-- INSERT --                                                    21,57          Top
```

2. 3.110.147.110 master | 3. 3.110.147.110 worker

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-deployment
  namespace: wanderlust
  labels:
    app: backend
spec:
  replicas: 1
  selector:
    matchLabels:
      app: backend
  template:
    metadata:
      name: backend
      labels:
        app: backend
    spec:
      containers:
        - name: backend
          image: premd91/wanderlust-backend-beta:latest
          ports:
            - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: backend-service
  namespace: wanderlust
-- INSERT --                                                    21,56          Top
```

Devops-Methodology






```
mongo-deployment-9d8749988-nghgn   1/1     Running   0          2m7s
redis-deployment-9fbf97b49-xxmjk   1/1     Running   0          5s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# vi frontend.yaml
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# vi backend.yaml
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# vi frontend.yaml
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f frontend.yaml
deployment.apps/frontend-deployment created
service/frontend-service created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl apply -f backend.yaml
deployment.apps/backend-deployment created
service/backend-service created
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

Best wishes from Devops Methodology

I have some error that's why the application is not accessing in port no 31000

What was the error

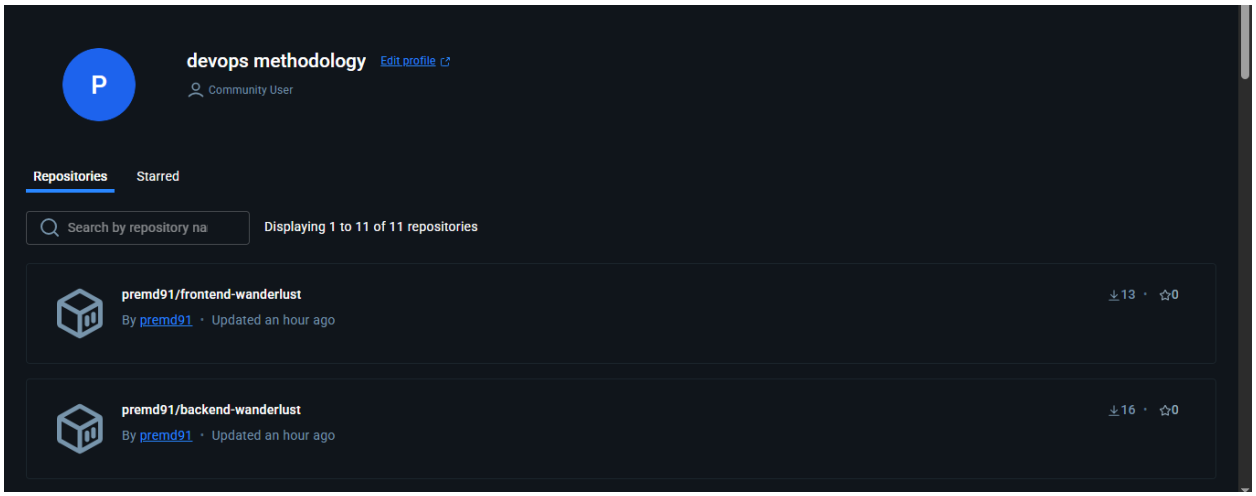
```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get all
NAME                                      READY   STATUS             RESTARTS   AGE
pod/backend-deployment-5cffb4b849-fwdj9   0/1     ImagePullBackOff   0          17s
pod/frontend-deployment-7f756db7b9-tmqzb  0/1     ErrImagePull       0          37s
pod/mongo-deployment-9d8749988-nghgn      1/1     Running            0          10m
pod/redis-deployment-9fbf97b49-xxmjk      1/1     Running            0          8m11s

NAME                       TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/backend-service    NodePort    10.97.195.48     <none>        8080:31100/TCP   17s
service/frontend-service   NodePort    10.108.248.71    <none>        5173:31000/TCP   37s
service/mongo-service      ClusterIP   10.102.92.89     <none>        27017/TCP        10m
service/redis-service      ClusterIP   10.109.120.127   <none>        6379/TCP         8m11s

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/backend-deployment    0/1     1            0           17s
deployment.apps/frontend-deployment   0/1     1            0           37s
deployment.apps/mongo-deployment      1/1     1            1           10m
deployment.apps/redis-deployment      1/1     1            1           8m11s

NAME                                            DESIRED   CURRENT   READY   AGE
replicaset.apps/backend-deployment-5cffb4b849   1         1         0       17s
replicaset.apps/frontend-deployment-7f756db7b9  1         1         0       37s
replicaset.apps/mongo-deployment-9d8749988      1         1         1       10m
replicaset.apps/redis-deployment-9fbf97b49      1         1         1       8m11s
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

Actually the image which I have build and tag it was not created the latest image for and in backend and frontend.yaml and I have updated as :latest which it can't pull the image as a devops we have to know the error but finally it was corrected

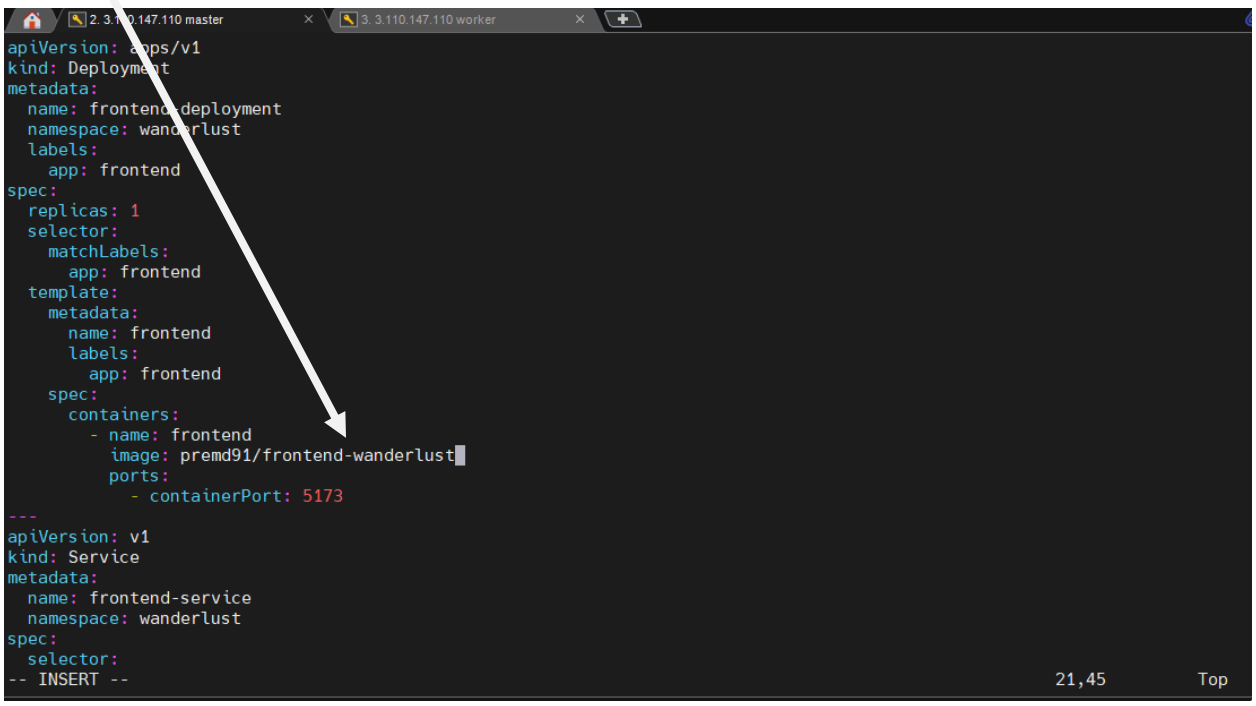
```
2. 3.110.147.110 master    3. 3.110.147.110 worker
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-deployment
  namespace: wanderlust
  labels:
    app: backend
spec:
  replicas: 1
  selector:
    matchLabels:
      app: backend
  template:
    metadata:
      name: backend
      labels:
        app: backend
    spec:
      containers:
        - name: backend
          image: premd91/wanderlust-backend-beta:latest
          ports:
            - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: backend-service
  namespace: wanderlust
spec:
  type: NodePort
"backend.yaml" 38L, 667B                                                    21,55
```

That's why the error was coming

But in dockerhub the latest was not added so I just eliminate it



Here's the correct format



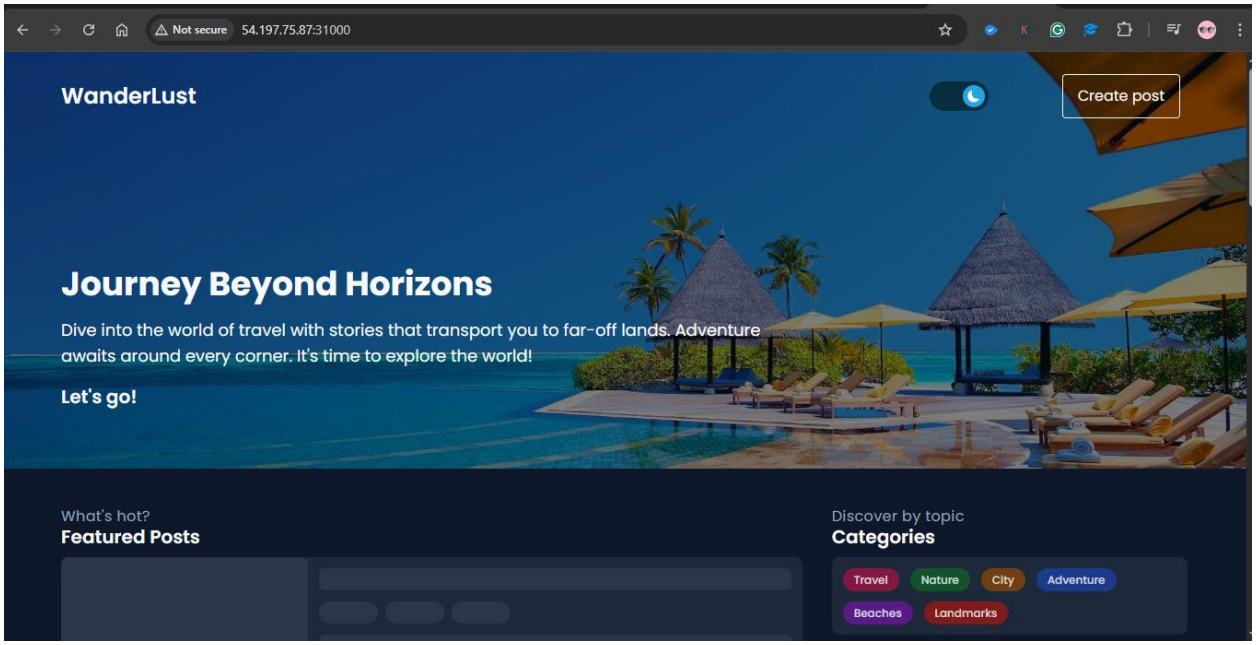
After that check my container is creating

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get all
NAME                                          READY   STATUS              RESTARTS   AGE
pod/backend-deployment-67b5d55b65-p5lk6       0/1     ContainerCreating   0          5s
pod/backend-deployment-c6994bb47-7wm9g        0/1     ImagePullBackOff    0          4m35s
pod/frontend-deployment-84dd648ccc-wsbq6      1/1     Running             0          15s
pod/mongo-deployment-9d8749988-nghgn          1/1     Running             0          32m
pod/redis-deployment-9fbf97b49-xxmjk          1/1     Running             0          30m

NAME                         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/backend-service      NodePort    10.97.195.48     <none>        8080:31100/TCP   22m
service/frontend-service     NodePort    10.108.248.71    <none>        5173:31000/TCP   22m
service/mongo-service        ClusterIP   10.102.92.89     <none>        27017/TCP        32m
service/redis-service        ClusterIP   10.109.120.127   <none>        6379/TCP         30m

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
```

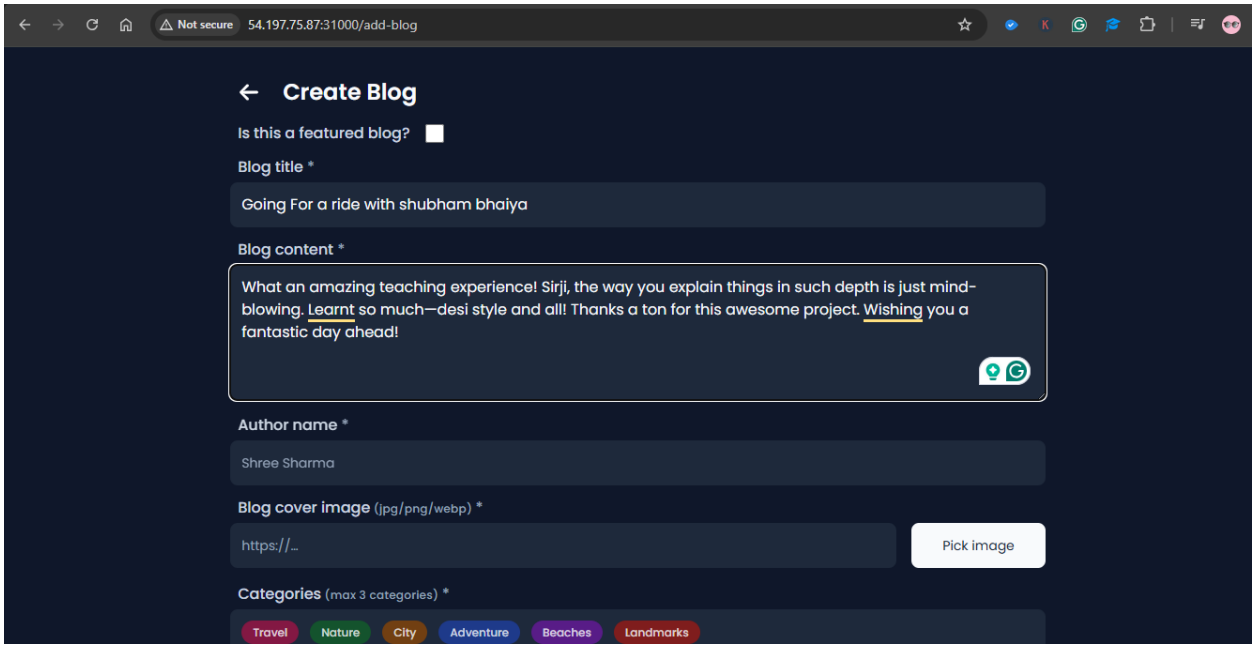
After sometime it shows the container was created so

```
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes# kubectl get all
NAME                                          READY   STATUS    RESTARTS   AGE
pod/backend-deployment-67b5d55b65-p5lk6       1/1     Running   0          30s
pod/frontend-deployment-84dd648ccc-wsbq6      1/1     Running   0          40s
pod/mongo-deployment-9d8749988-nghgn          1/1     Running   0          33m
pod/redis-deployment-9fbf97b49-xxmjk          1/1     Running   0          30m

NAME                         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/backend-service      NodePort    10.97.195.48     <none>        8080:31100/TCP   23m
service/frontend-service     NodePort    10.108.248.71    <none>        5173:31000/TCP   23m
service/mongo-service        ClusterIP   10.102.92.89     <none>        27017/TCP        33m
service/redis-service        ClusterIP   10.109.120.127   <none>        6379/TCP         30m

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/backend-deployment       1/1     1            1           23m
deployment.apps/frontend-deployment      1/1     1            1           23m
deployment.apps/mongo-deployment         1/1     1            1           33m
deployment.apps/redis-deployment         1/1     1            1           30m

NAME                                                DESIRED   CURRENT   READY   AGE
replicaset.apps/backend-deployment-54f669f69d       0         0         0       11m
replicaset.apps/backend-deployment-5cffb4b849       0         0         0       23m
replicaset.apps/backend-deployment-67b5d55b65       1         1         1       30s
replicaset.apps/backend-deployment-c6994bb47        0         0         0       5m
replicaset.apps/frontend-deployment-7f6f849fdc      0         0         0       5m12s
replicaset.apps/frontend-deployment-7f756db7b9      0         0         0       23m
replicaset.apps/frontend-deployment-84dd648ccc      1         1         1       40s
replicaset.apps/frontend-deployment-9f7d74586       0         0         0       12m
replicaset.apps/mongo-deployment-9d8749988          1         1         1       33m
replicaset.apps/redis-deployment-9fbf97b49          1         1         1       30m
root@ip-172-31-84-134:/home/ubuntu/wanderlust/kubernetes#
```

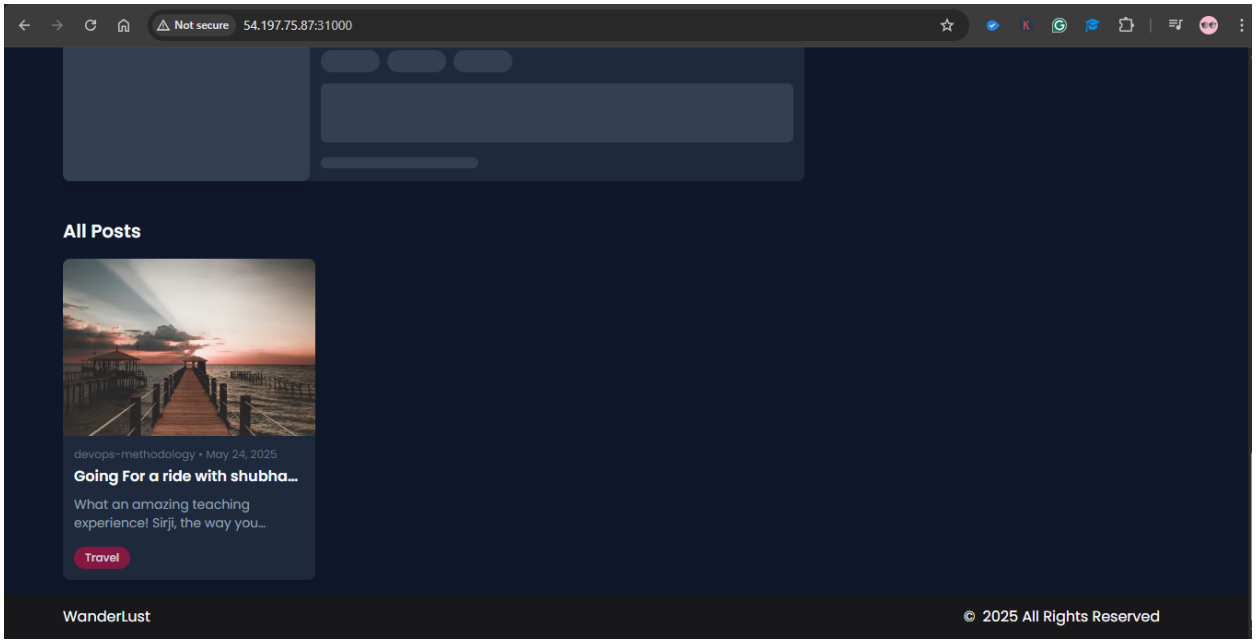
I have just access workernode ip:31000(for the frontend)

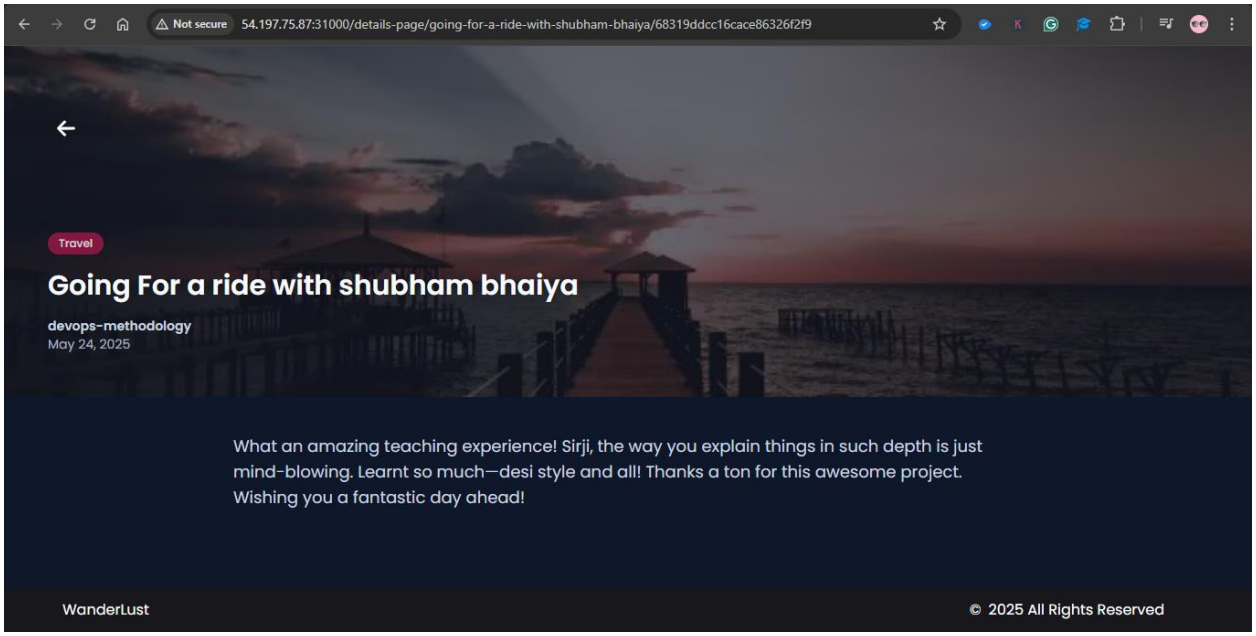


Also create the blog for devops wale bhaiya urff shubham bhaiya



Devops-Methodology





Best wishes from Devops Methodology

Workernodeip:31100(for backend)

Yay!! Backend of wanderlust app is now accessible